Database Design Project
CS 3200 - Fall 2025

Part 1: Team and Project Info

Team Name:	Core Four
Product/Project Name:	FitFlow

Description	**Link**
Link to shared copy of this exact Google Doc	https://docs.google.com/document/d/1JwE7iO30bSZjNgv0rCFN2CASjYY9RfzKJY_J8d62DMk/edit?usp=sharing
Github Project Repo Link	https://github.com/ranya1958/FitFlow.git
Link to shared demo video (guidelines in Project Phase 3 Document)*	https://www.youtube.com/watch?v=kRvscEVxQec

Part 2: Elevator Pitch of Project/Product

FitFlow is a health and fitness tracking website meant to connect gym members, trainers, and health professionals in one platform. Administrators of the application manage accounts and oversee performance data, while trainers and health analysts will monitor member progress and design personalized programs. Student members can log workouts, view trends, and track their growth. Essentially, FitFlow creates a connected ecosystem that turns health data into actionable insights. Continuously it is an essential app to bridging the gap between online trainers and members to effectively track their progress and identify who needs follow-up or motivation, assign workout programs to specific members and see client progress metrics so they can adjust their programs appropriately. Prior to FitFlow, trainers relied on Excel spreadsheets and text messages to manage clients which often led to a loss of data and made it difficult to monitor client consistency. Now, trainers have a centralized platform to program workouts, track clients, and maintain data integrity leading to calculated coaching decisions.

Part 3: User Personas

<u>Persona 1: Ava Martinez</u>

What's Their Story:

Ava Martinez is a 27 year old Systems Manager based in Boston, MA who manages the backend infrastructure for a fast-growing fitness platform, FitFlow. She works behind the scenes to ensure that the exercise library, trainer dashboards, and user data stay accurate, secure, and up-to-date.
Until now, Ava relied on a patchwork of spreadsheets and manual tracking, which made it difficult to user reports and catch errors in trainer uploads, fix broken video links, or remove outdated exercises. She also struggled to monitor system performance, manage user and trainer accounts, and maintain data integrity across multiple disconnected tools. She also deals with fraudulent accounts or spam content, and often lacks visibility into what's breaking when users report bugs or system errors. Ava needs a centralized management platform that streamlines these workflows. With these tools, Ava can ensure FitFlow remains reliable, organized, and professional for trainers and clients, while saving time and reducing errors in her workflow.

User Stories for Persona 1:

1. ~~As a system administrator, I want to add new trainer or trainee profiles so that new users can start using the platform and accessing exercises immediately. (Add new data)~~ ***(Split into two different stories)***
    a. ***Modification: As a system administrator, I want to add a user for a trainer profile that is verified with certifications so that new trainers can start using the platform immediately.***
2. ***Addition: As a system administrator, I want to add a user for a client profile so that new trainees can access their workouts right away.***

~~3. As a system administrator, I want to add new exercises submitted by verified trainers so that the exercise library stays comprehensive and up-to-date. (Add new data)~~ ***Deleted***

3. As a system administrator, I want to remove outdated or incorrect exercises so that the platform stays clean and organized. (Remove data)
4. ~~As a system administrator, I want to manage trainer accounts and permissions so that only qualified trainers can modify the exercise library.~~
    a. **Modification: As a system administrator, I want to manage all user accounts/profiles and permissions so that each role has the correct level of access to the platform's features.**
5. As a system administrator, I want to review flagged exercises or reported issues so that inappropriate or incorrect content is corrected or removed.
6. As a system administrator, I want to monitor system logs and performance dashboards so that issues can be identified and resolved before they affect users.
7. As a system administrator, I want the database to log when backups were last performed and notify me when a new one is due, so that data safety is maintained consistently.

Persona 2: Marcus Rodriguez

What's Their Story?

Marcus Rodriguez is a 32-year-old certified personal trainer located in Boston, MA who recently enhanced his business to online after several years of in-person training. He currently manages 15–20 clients online and is trying to improve his approach to how he delivers workouts and tracks client progress. Until now, he relied on Excel sheets and text messages, which led to a loss of data, limited insight on whether clients actually completed their workouts, difficulty adjusting programs based on client feedback, and he often lost clients because he seemed "disorganized" or "not tech-savvy". He's not really technical but wants a tool that saves him time, looks more professional, allows him to create reusable workout program templates and assign specific ones to clients, view client workout completion rates in real-time and add feedback/notes on client workouts. And track which clients are falling behind/at-risk clients to modify programs on the fly based on client performance.

User Stories for Persona 2:

1. As a personal trainer, I need to create workout programs with exercises, sets, reps, and rest periods so that I can reuse them and create effective programs for each client.
2. As a personal trainer, I need to organize and assign specific workout programs for clients so that they know exactly what to do each week.
3. As a personal trainer, I need to view which clients have completed their assigned workouts so that I can identify who needs follow-up or motivation.
4. As a personal trainer, I need to see client progress metrics like consistency rate or personal records so that I can adjust their programs appropriately in the case it's too difficult or they need more challenges.
5. ~~As a personal trainer, I need to be able to update clients' new personal records so that I can adjust for growth and new goals.~~
    a. **Modification: As a personal trainer, I need to review clients' updated personal records so that I can adjust their future workout programs appropriately.**
6. ~~As a personal trainer, I need to be able to leave notes/feedback on client workout logs so that I can provide encouragement or form corrections.~~
    a. Modification: **As a personal trainer, I need to be able to leave notes/feedback on client workout logs so that I can provide encouragement or form corrections or adjust for any injuries, changes in health, or changing goals.**

Persona 3: Dr. Kennedy Smith

What's their Story:

Dr. Kennedy Smith is a 36-year-old doctor with a PhD in exercise science working as a Health Analyst for the fitness platform company and is responsible for understanding user behaviors and health. She collects and analyzes data needed to guide product development and business decisions to report to executives on user retention and engagement. Dr. Smith struggles with identifying why users stop using the app, features that promote long-term engagement, and patterns in successful versus unsuccessful members. Furthermore, Dr. Smith lacks the visual analysis for trainer effectiveness and member's health outcomes. Through FitFlow the Health Analyst is able to discover health trends, retention patterns, and efficiency program dynamics to promote the longevity of the user experience.

User Stories for Persona 3:

1. ~~As a health analyst, I need workout trends, durations, and patterns of all members individually and as an average.~~
    a. **Modification: As a health analyst, I need the workout duration of all members individually and as an average based on completion status.**
2. As a health analyst, I need client background information including demographics, fitness levels, and goals.
3. ~~As a health analyst, I need analytical dashboards composed of workout/trainer retention curves and trainer program cohort analysis.~~
    a. **Modification: As a health analyst, I need each client's most recent health metric record to track their latest status.**
4. ~~As a health analyst, I need analytics on exercise and gym feature popularity along with health progression/decline.~~
    a. **Modification: As a health analyst, I need analytics on health progression or decline based on weight and body fat percentage.**
5. ~~As a health analyst, I need workout completion rates and durations related to program succession analysis.~~
    a. **Modification: As a health analyst, I need workout completion rates related to each program to track succession analysis.**
6. ~~As a health analyst, I need analytic dashboards of trained versus untrained member metrics and health progression/retention.~~
    a. **Modification: As a health analyst, I need to find which workout templates are most frequently used by the clients.**

Persona 4: Chester Stone

What's Their Story:

Chester Stone, 20, is a sophomore at Northeastern majoring in Business Administration. He started lifting at Marino Center 8 months ago and has gained 15 pounds, but he's hit a plateau. He works 10 hours/week on campus in the library, and takes a full load of courses in preparation for co-op. His workout times are inconsistent (7am or 9pm depending on his schedule), and he frequently forgets previous workout details, relying on random TikTok plans saved in his camera roll that he can't find when needed. His main frustrations: can't remember previous performance, no clear progression plan, can't see progress, loses motivation during midterms and weekend transitions, no accountability system, and feels intimidated during peak gym hours due to uncertainty about proper form. He needs a mobile app that tracks lifts ~~automatically~~, shows what to do each session with progression built in, and uses ~~streak~~ completion tracking to maintain consistency despite his chaotic student schedule.

User Stories for Persona 2:

1. ~~As a client, I want to quickly log my completed sets (exercise, weight, reps) during my workout so that I don't waste time at the gym and can get to my next class on time.~~
    a. **Modification: As a client, I want to log my completed workouts with duration and notes so that I can track what I accomplished each session.**
2. ~~As a client, I want to see what weight and reps I used for each exercise in my last workout so that I can lift slightly more today and make consistent progress.~~
    a. **Modification: As a client, I want to view my most recent completed workouts so that I can see which sessions I've done and stay motivated.**
3. ~~As a client, I want to browse beginner-friendly workout programs and follow one so that I stop doing random exercises and actually follow a structured plan.~~
    a. **Modification: As a client, I want to view my coach-assigned weekly program with all exercise details so that I know exactly what to do each training session.**
4. ~~As a client, I want the app to automatically detect and celebrate when I hit a new personal record so that I stay motivated during tough weeks like midterms.~~
    a. **Modification: As a client, I want to see how many workouts I've completed this month compared to last month so that I can track if my consistency is improving.**
5. ~~As a client, I want to see my current workout streak and calendar history so that I feel accountable and don't want to break my streak during stressful periods.~~
    a. **Modification: As a client, I want to see how many personal records I've achieved this month so that I can track my progress and celebrate my improvements."**
6. ~~As a client, I want to edit or delete workout entries so that I can fix mistakes like typos or remove warmup sets that don't represent my actual training.~~

a. **Modification:** 4.6) As a client, I want to delete incomplete workout logs so that my workout history only shows actual training sessions I completed.

Part 3: Wireframes

<u>Wireframe 1: System Administrator</u>

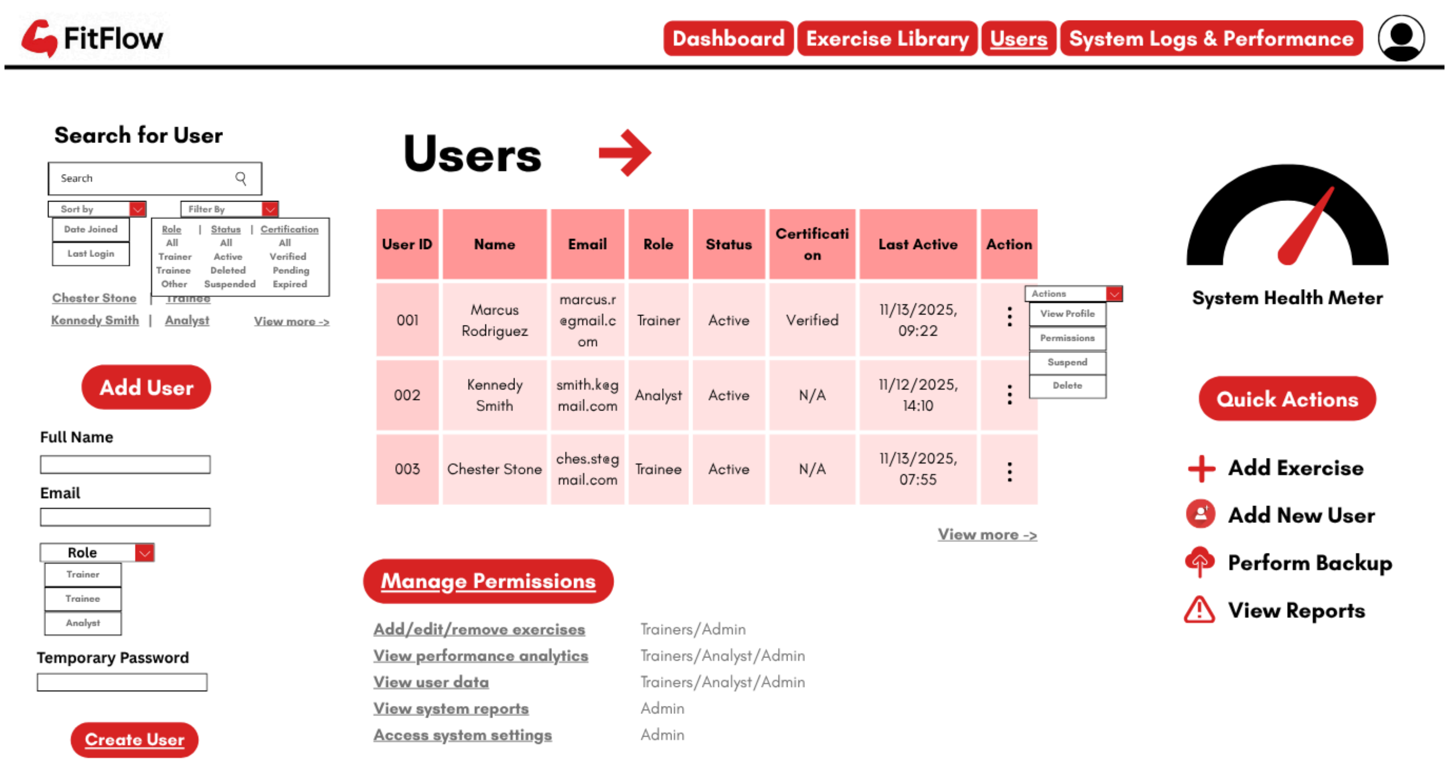
Intended user persona: <u>Ava Martinez</u>
Intended user stories: 1, 5, 6, & 7

This screen serves as Ava's main dashboard where she can gain an overview of FitFlow's current activity. It summarizes all essential metrics such as the total number of trainers, trainees, exercises, flagged items, and the estimated time until the next database backup, directly supporting user story 7 by showing when backups were last completed and when the next one is due. The "Activity Feed" provides a chronological record of recent system events, exercises added, backup and user updates, and flagged content for review, aligning with the monitoring needs described in user story 6. The dashboard also includes a System Health Meter, reinforcing user story 6 by allowing Ava to instantly assess platform stability and detect issues early. Quick-access action buttons enable her to immediately add new exercises (user story 2), add users (user story 1), manage permissions (user story 4), respond to flagged items (user story 5), or initiate a manual backup (user story 7).

Wireframe 2: System Administrator

Intended user persona: Ava Martinez
Intended user stories: 1, 4, & 5

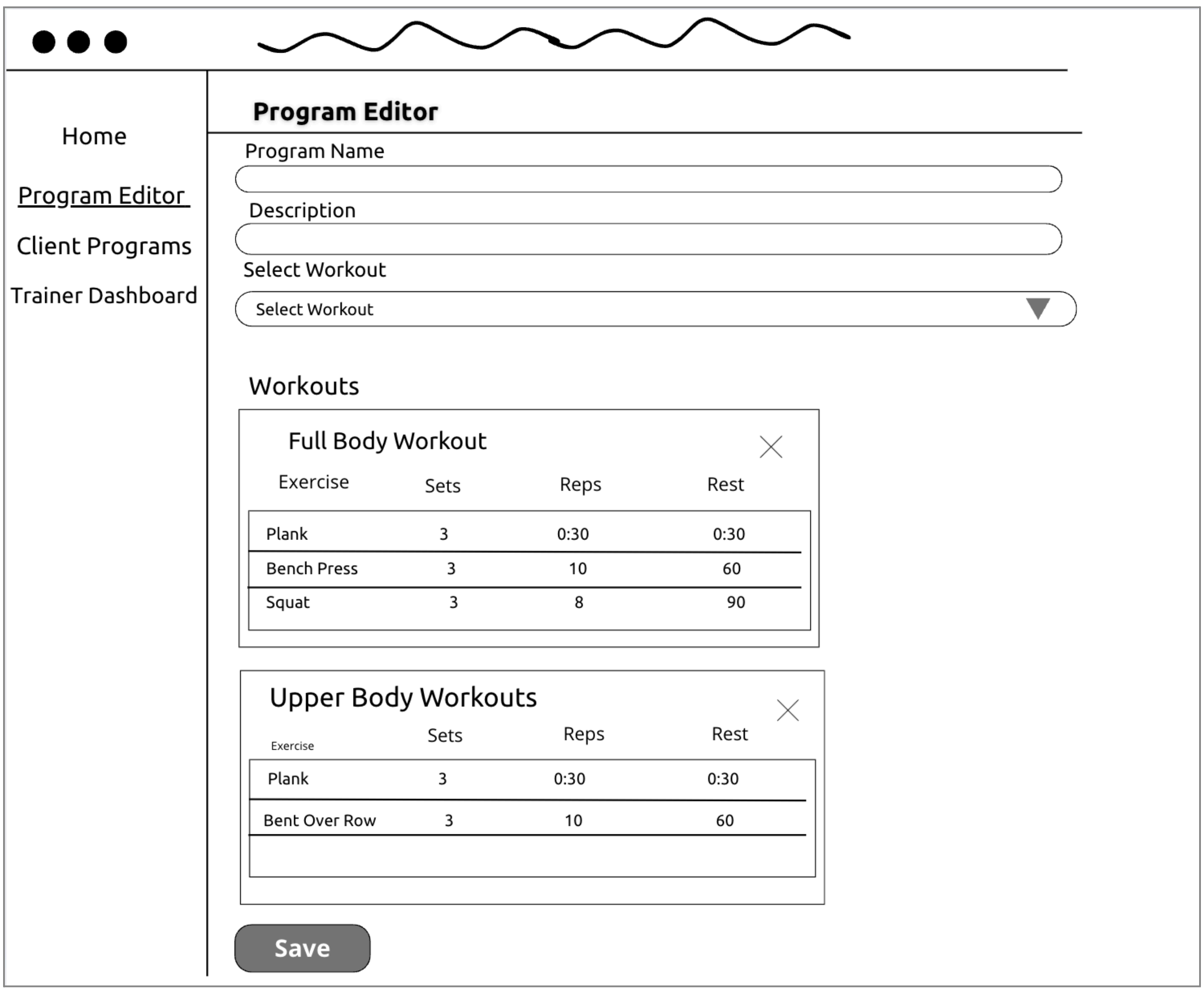
This screen allows Ava to manage all trainer, trainee and other accounts in a centralized, organized layout. The user table displays essential information, including name, email, role, certification status, and account activity, supporting user story 1, to allow Ava to add new profiles by using the "Add New User" button, edit existing user details, or deactivate outdated or fraudulent accounts.The page also enables Ava to adjust user permissions through dedicated controls, fulfilling user story 4. Integrated filters and search functionality help her quickly locate specific users, while indicators such as certification badges or "needs verification" flags give her instant visibility into account status. It also supports user story 5 by surfacing user accounts connected to flagged content, allowing Ava to review or restrict them if needed.



Wireframe 3: Trainer

Intended user persona: Marcus Rodriguez
Intended user stories: 1 & 5

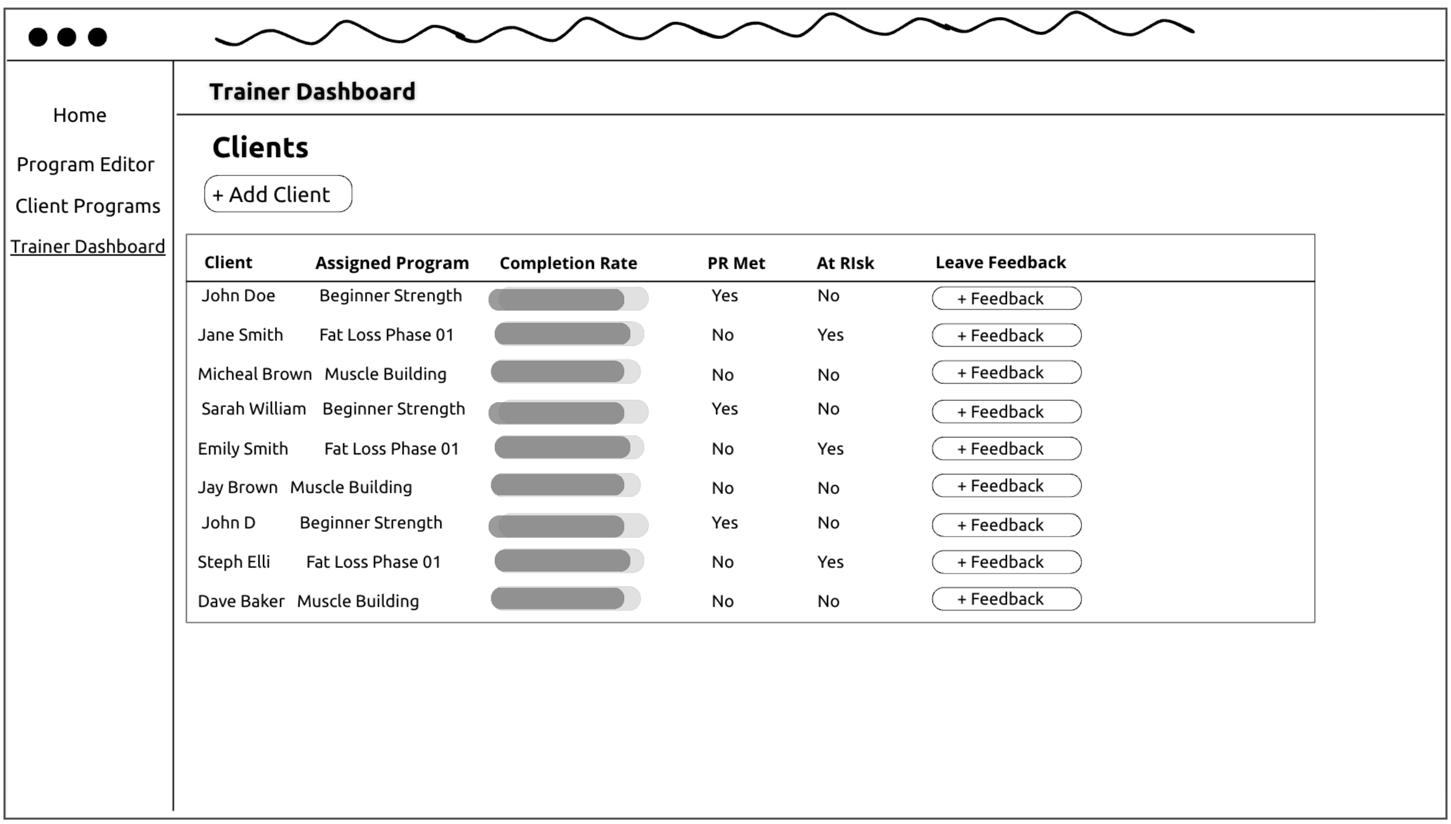
This screen serves as Marcus' Program Editor where he can create or modify workout programs for individual clients or templates for reuse efficiently. It's structured to make adding exercises fast and simple. This wireframe addresses his user story 1, of creating reusable workout program templates, and 5 being able to modify assigned programs for clients to adjust to any injuries, PR, or challenges.



Wireframe 4: Trainer

Intended user persona: Marcus Rodriguez
Intended user stories: 3 & 4

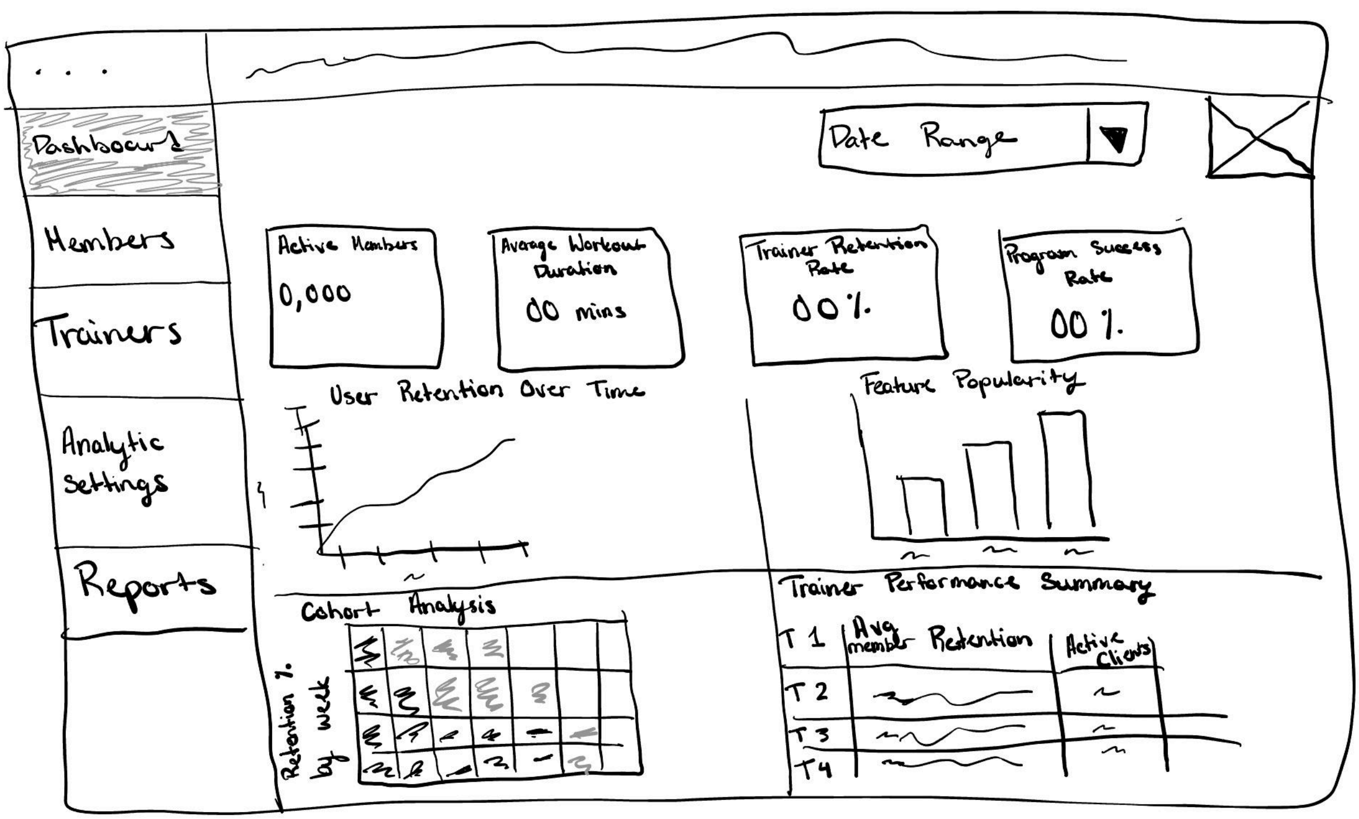
This screen serves as Marcus' Trainer Dashboard where he can input and view all his clients, their programs, and completion rate. It provides him with a quick organized visual summary of client progress; who's on track or falling behind. This wireframe addresses his user story 3 as he can see which clients have completed their assigned workouts so that he can identify who needs follow-up or motivation, and user story 6 as he is able to leave notes/feedback on client workout logs to provide encouragement or form corrections.



Wireframe 5: Health Analyst

Intended user persona: Dr. Kennedy Smith
Intended user stories: 1 & 3

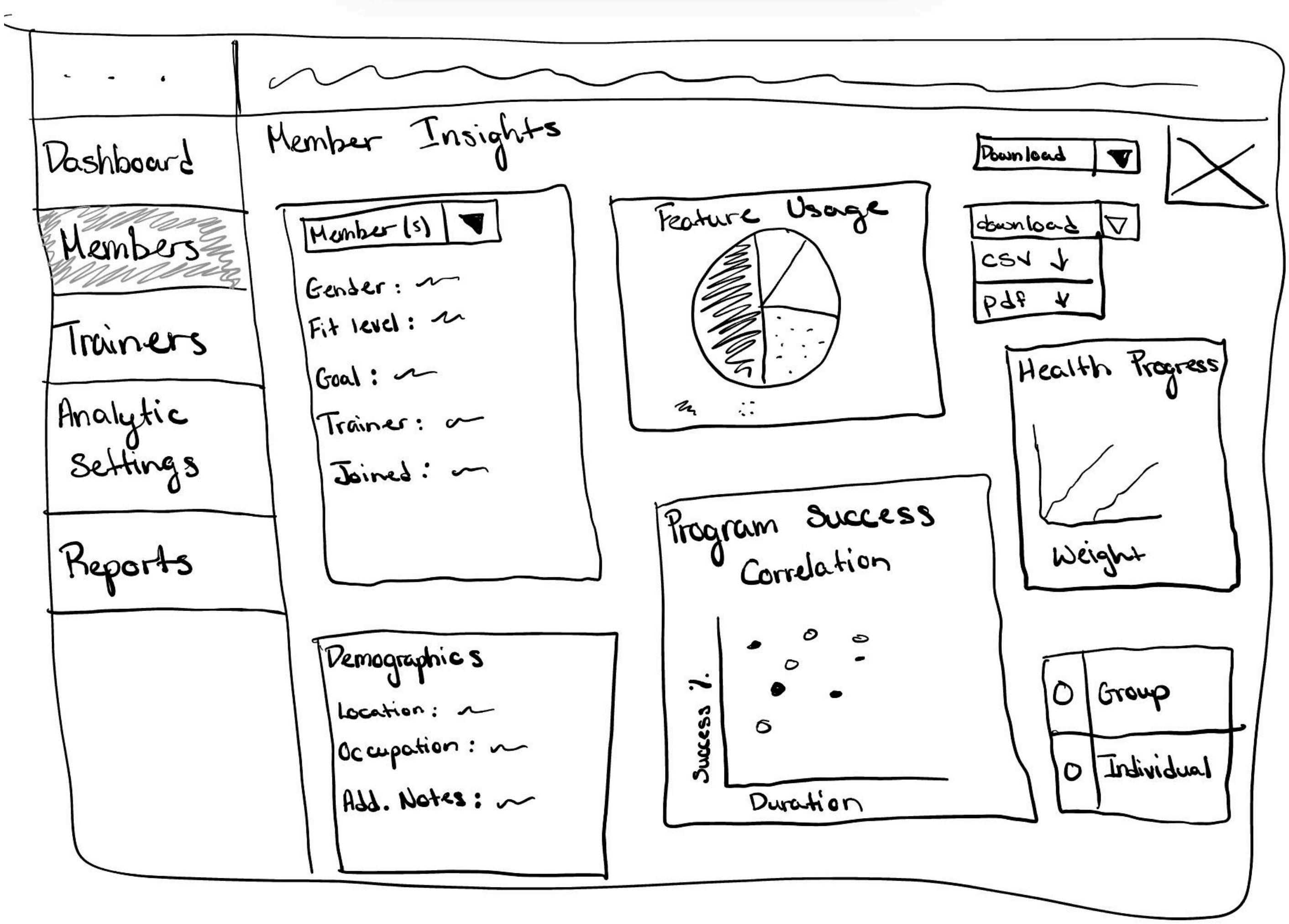
This wireframe is a Health Analyst Dashboard Overview geared to give Dr.Smith a glance of user engagement, health trends, and trainer effectiveness. The top header displays the brand logo in the left hand corner, profile icon on the right upper corner, date filter. The side bar to the left of the screen demonstrates tabs for the dashboard as the current screen, members, trainers, analytics settings, and reports. The main area of the dashboards is split into three parts with the first section containing key metrics, calculating active members, average workout durations supporting user story 1, trainer retention rate which supports user story 3, and program success rate. The second section displays graphs geared towards the correlation of engagement trends, being user retention and feature popularity. The third section visualizes a trainer program effectiveness and summaries through a heatmap and summary table.



Wireframe 6: Health Analyst

Intended user persona: Dr. Kennedy Smith
Intended user stories: 2, 4, & 5

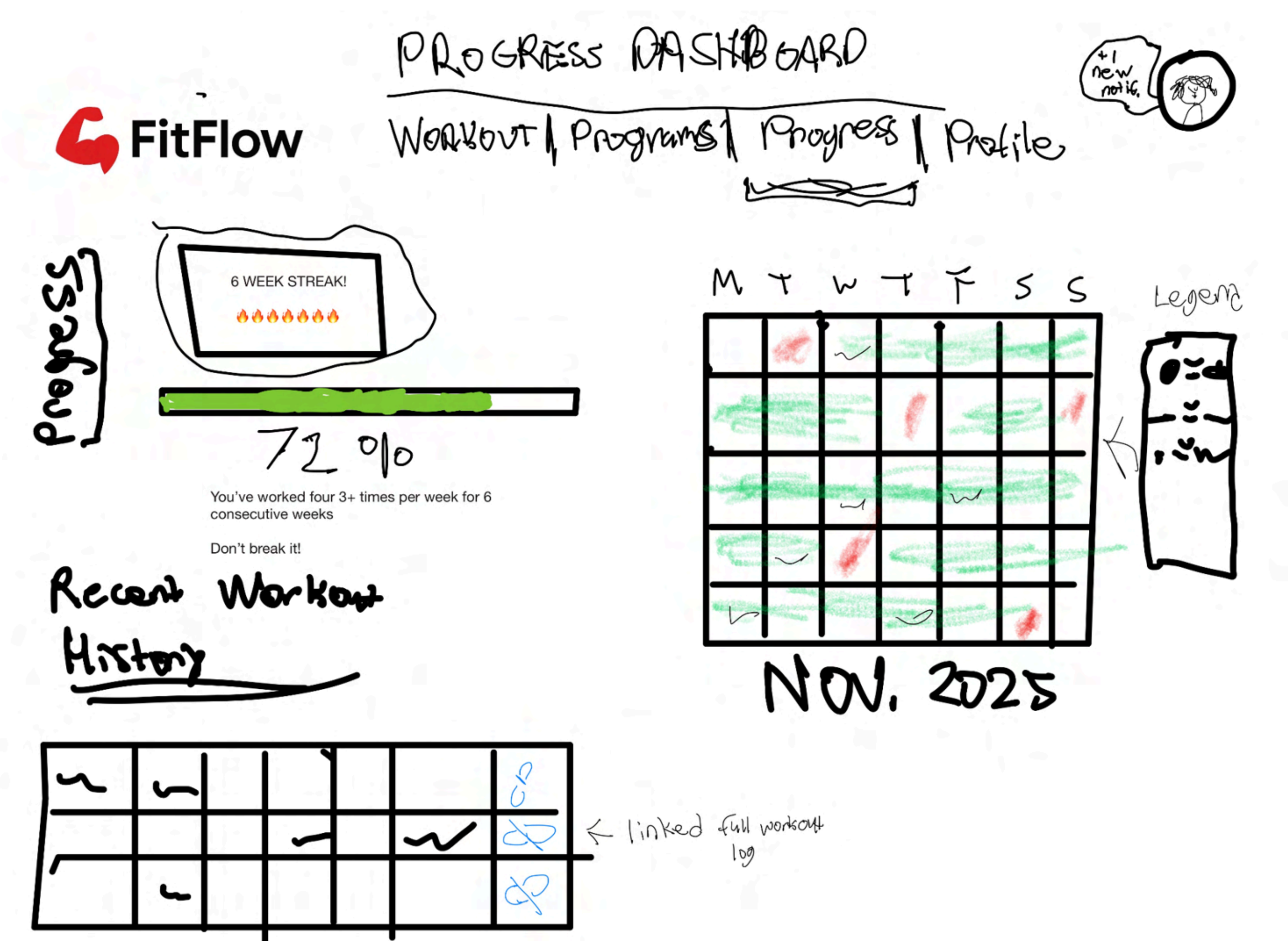
This wireframe represents the Member Health and Behavior Analysis Page which is geared to dissect individual or various grouped member health trends, background data, and success patterns. The top header of the wireframe includes the title of the page along with filters for the trainer, program, time range, and fitness level. Lastly, the header has an export button with the ability to transform the data into a portable csv or pdf file. The main portion of the wireframe will be split into two sections, the first one containing the member profile summary and the second including the analytical visual of the chosen member. The Member Profile Summary includes general information about the member along with residence, occupation, and additional health notes which supports user story 2. The section with analytic visuals include a chart to demonstrate workout completion and one to visualize health progress/BMI. The visuals also include a pie chart to track feature engagement to support user story 4 and a scatter plot for program success correlation by trainer demonstrating user story 5.



Wireframe 7: Client

Intended user persona: Chester Stone
Intended user stories: 4 & 5

This screen serves as Chester's Progress Dashboard where he can view his workout consistency, personal records, and training history. This wireframe addresses his user story #4 as the app automatically detects and celebrates when he hits new personal records in the dedicated PR section, keeping him motivated during tough weeks like midterms, and user story #5 as he can see his current workout streak prominently displayed at the top along with a calendar heatmap showing his complete workout history for the month, so he feels accountable and doesn't want to break his streak during stressful periods.



Wireframe 8: Client

Intended user persona: Chester Stone
Intended user stories: 1, 2, & 6

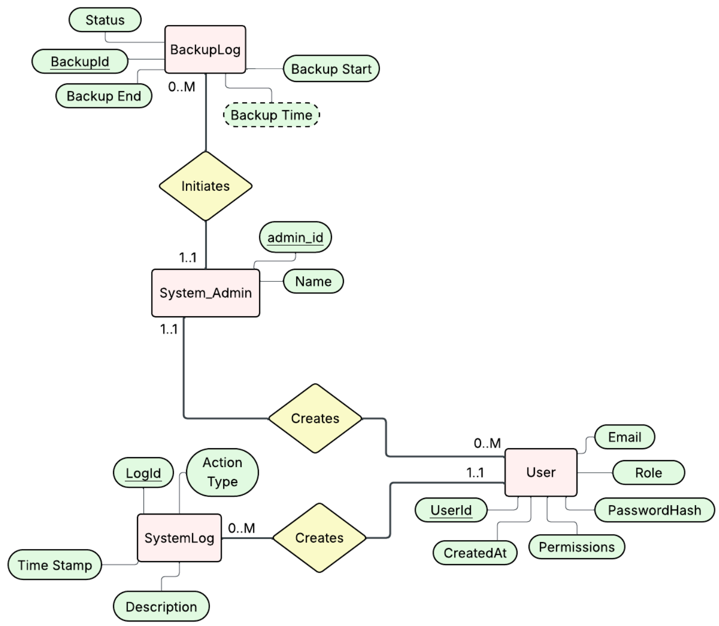
This screen serves as Chester's Active Workout Logging interface where he can quickly log his sets in real-time during his gym session. This wireframe addresses his user story #1 as he can quickly log completed sets (exercise, weight, reps) without wasting time at the gym between classes, user story #2 as he can see what weight and reps he used in his last workout displayed prominently at the top so he knows what to aim for today, and user story #6 as he can edit or delete workout entries using the action icons next to each completed set to fix typos or remove warmup sets that don't represent his actual training.

Part 4: Localized Conceptual Data Models as ER Diagrams

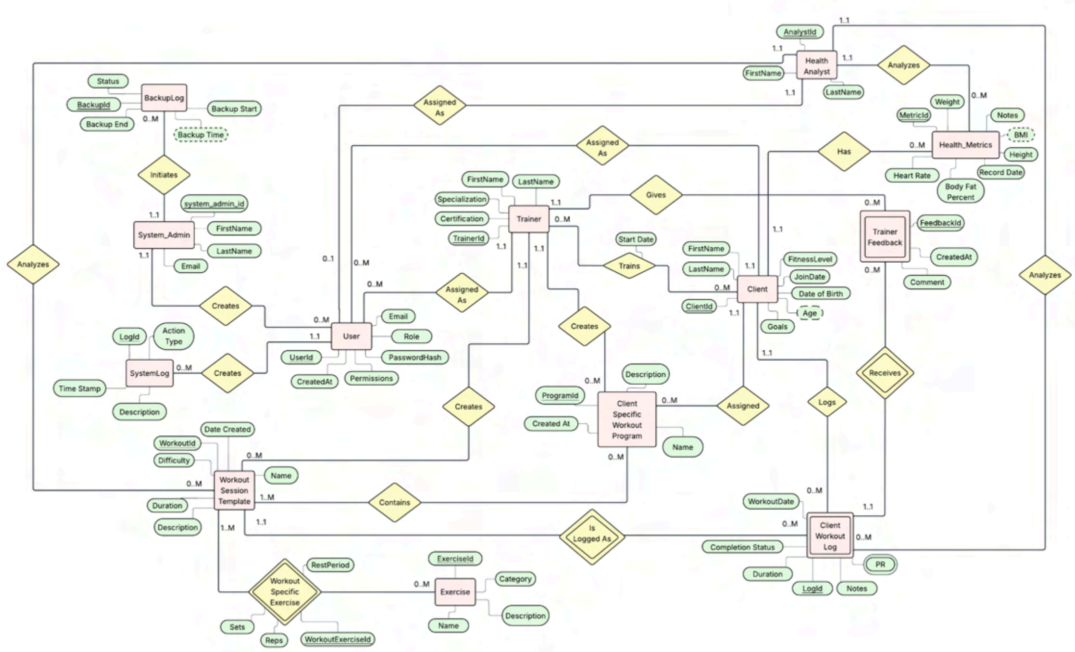
User Persona 1: Ava Martinez - System Admin

(Also has access to entire database, so Global ERD applies as well)

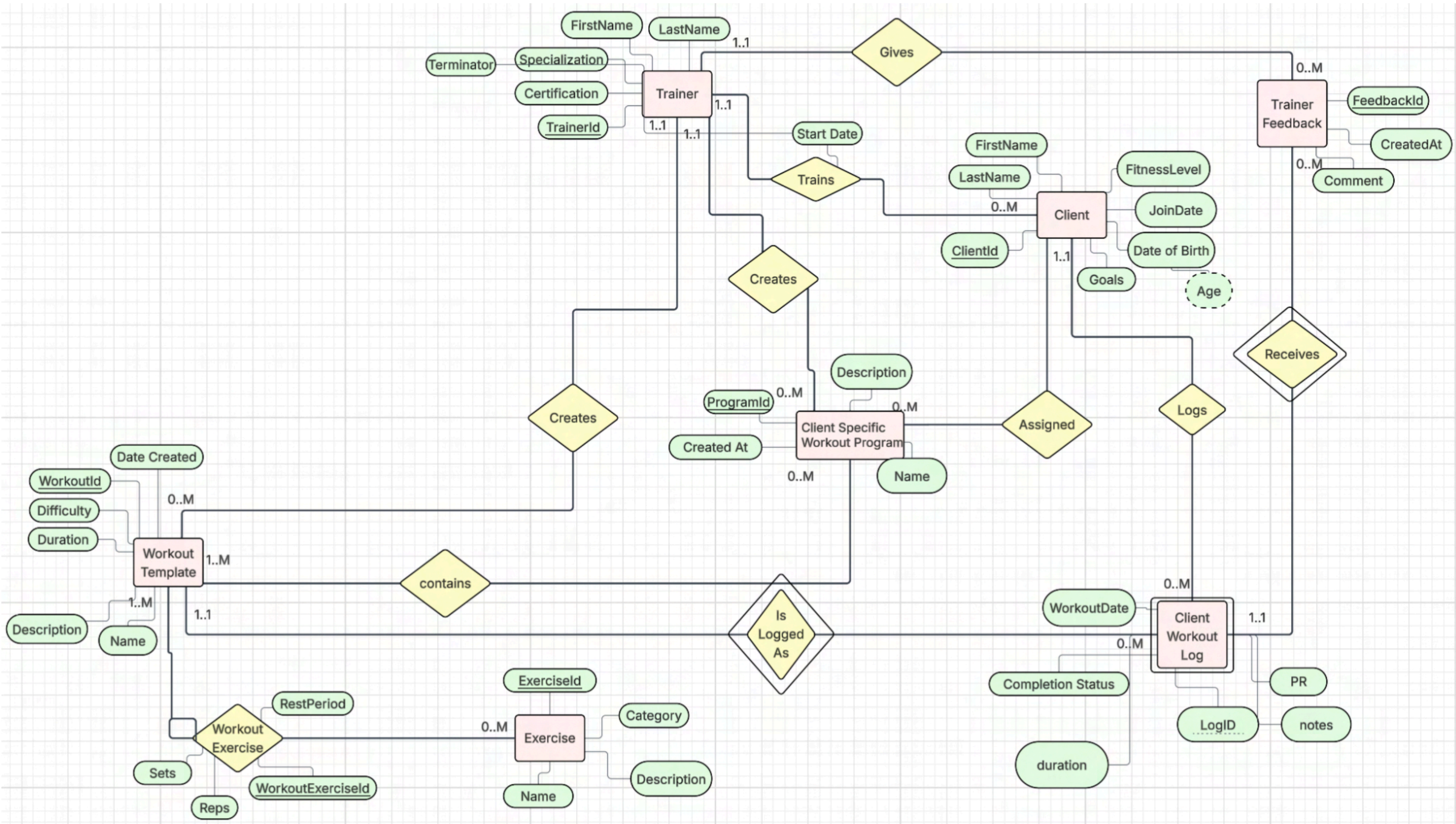
ERD for Main Functions:



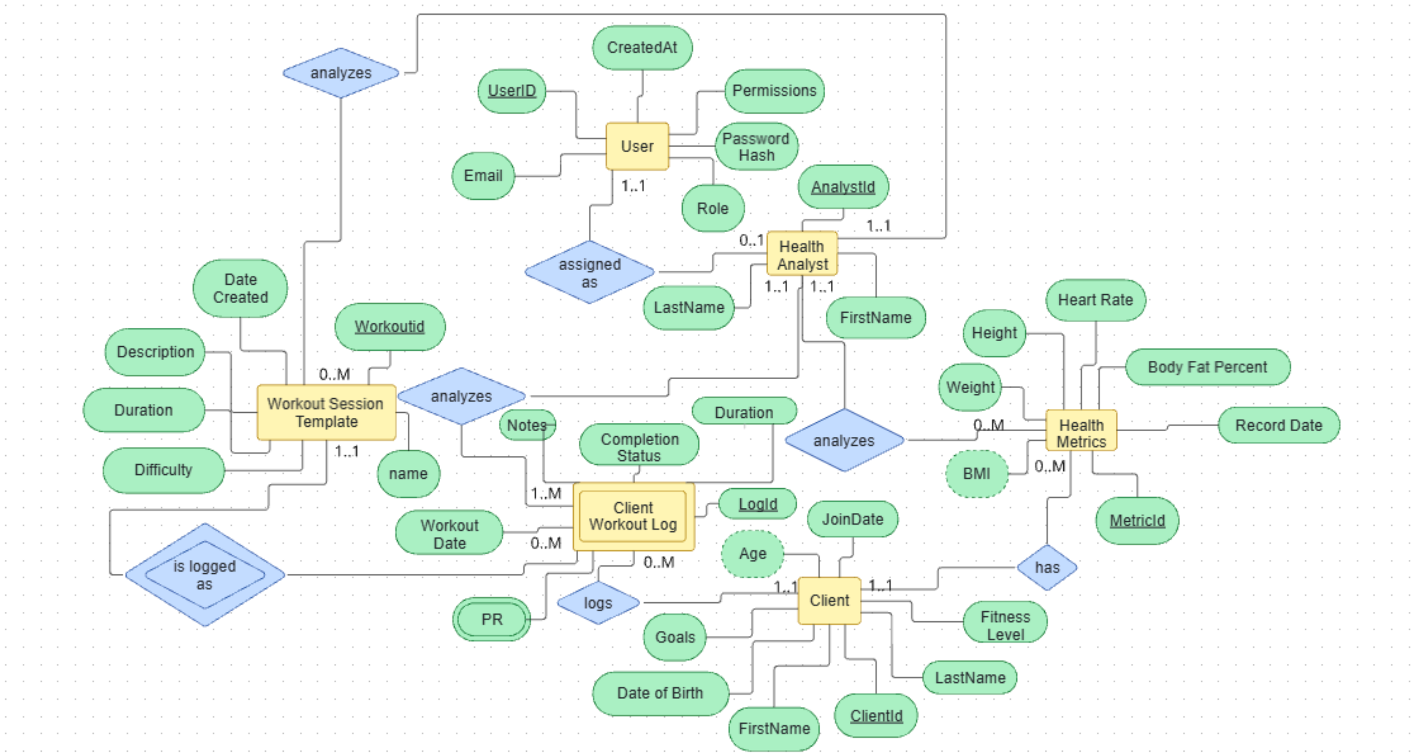
Global ERD



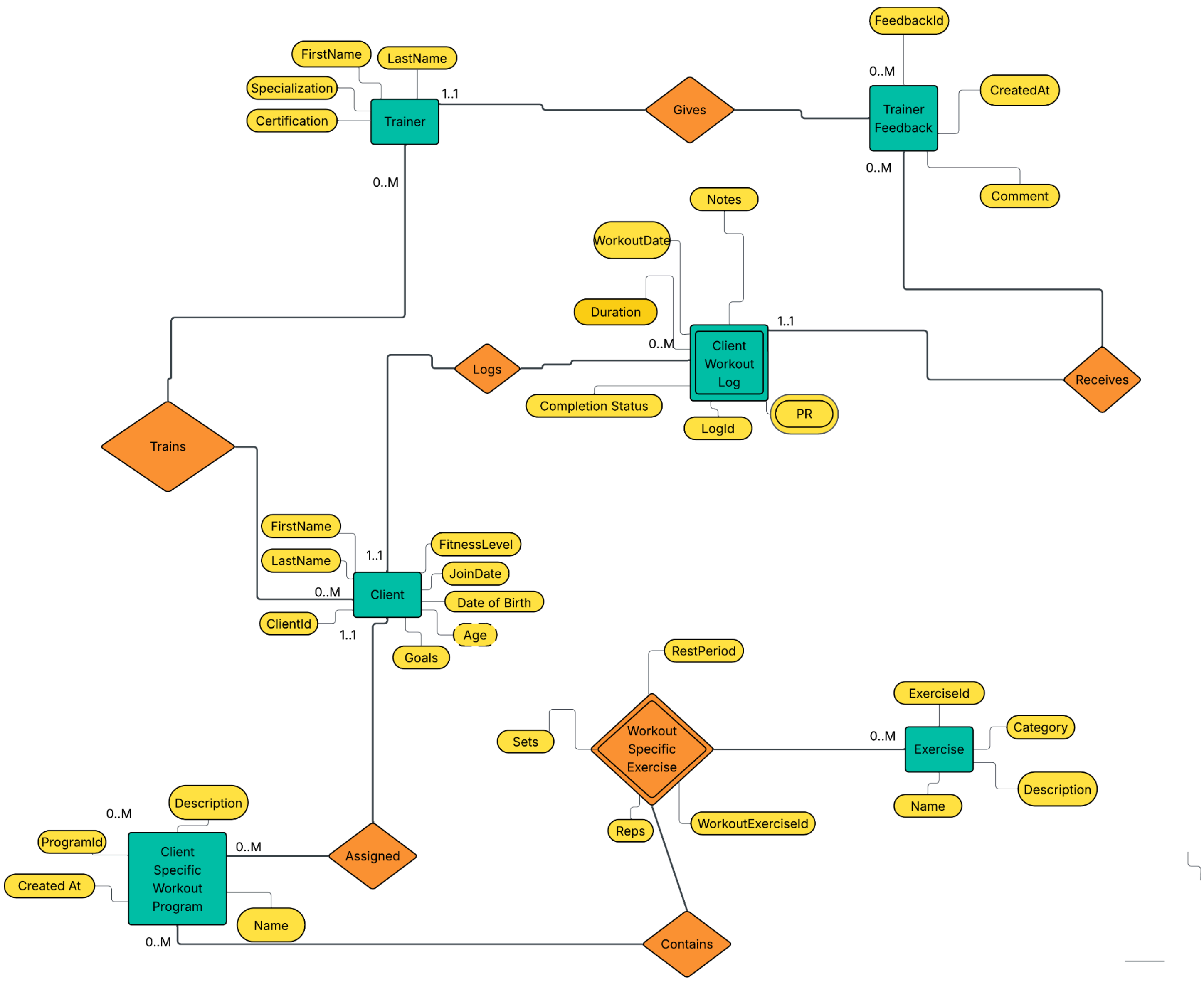
User Persona 2: Marcus Rodriguez - Trainer



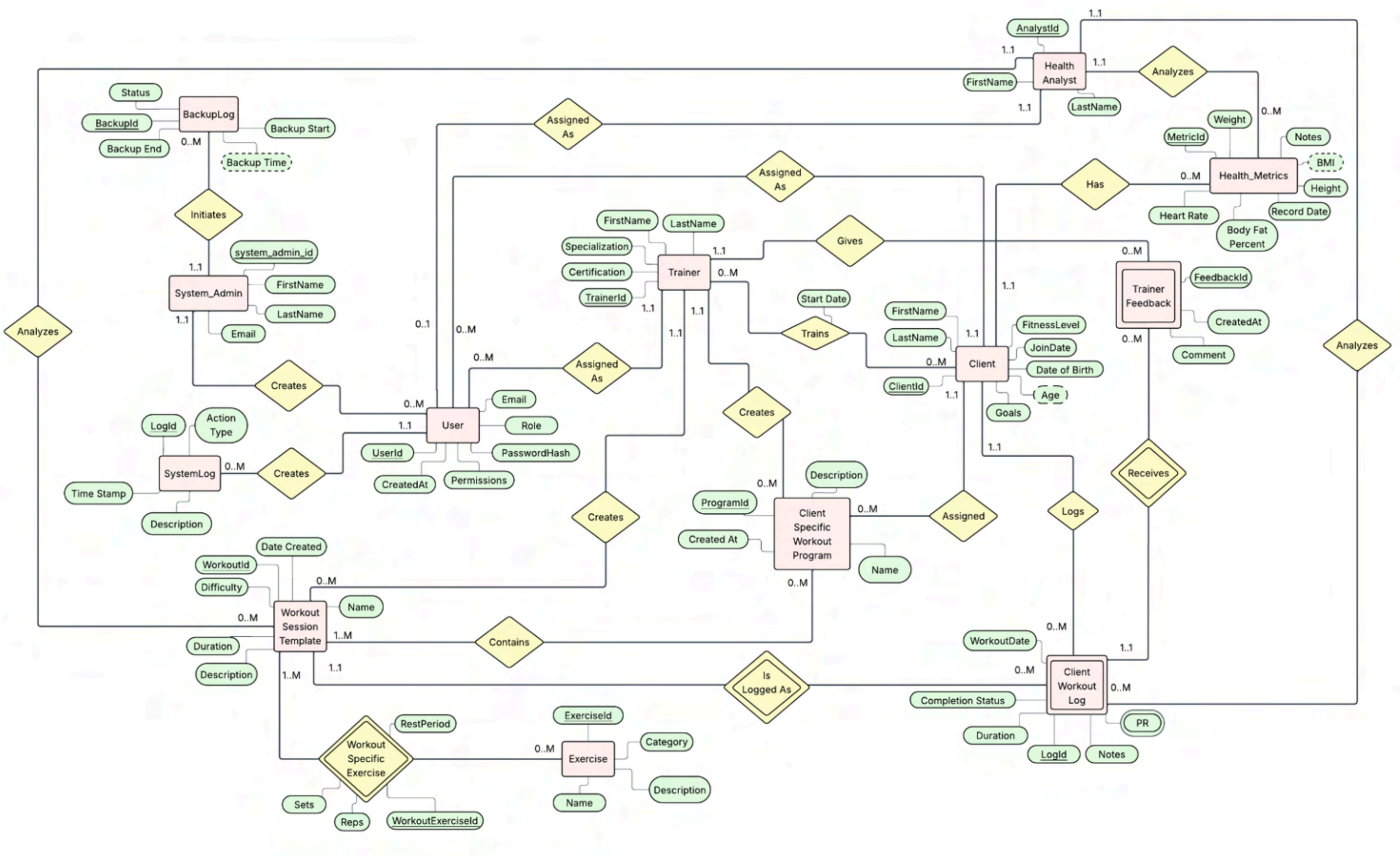
<u>User Persona 3: Dr. Kennedy Smith - Health Analyst</u>



User Persona 4: Chester Stone - Client

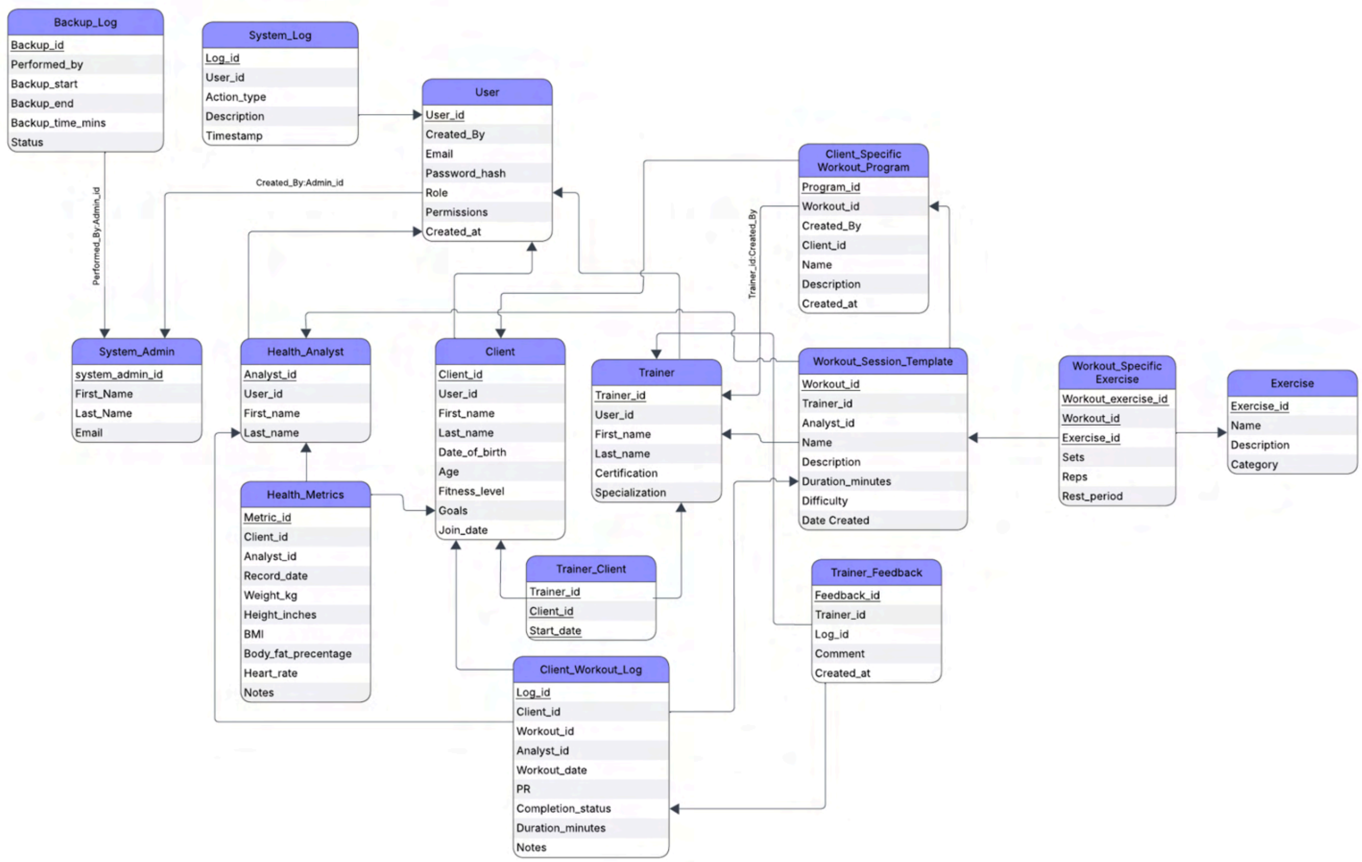


Global ER Diagram



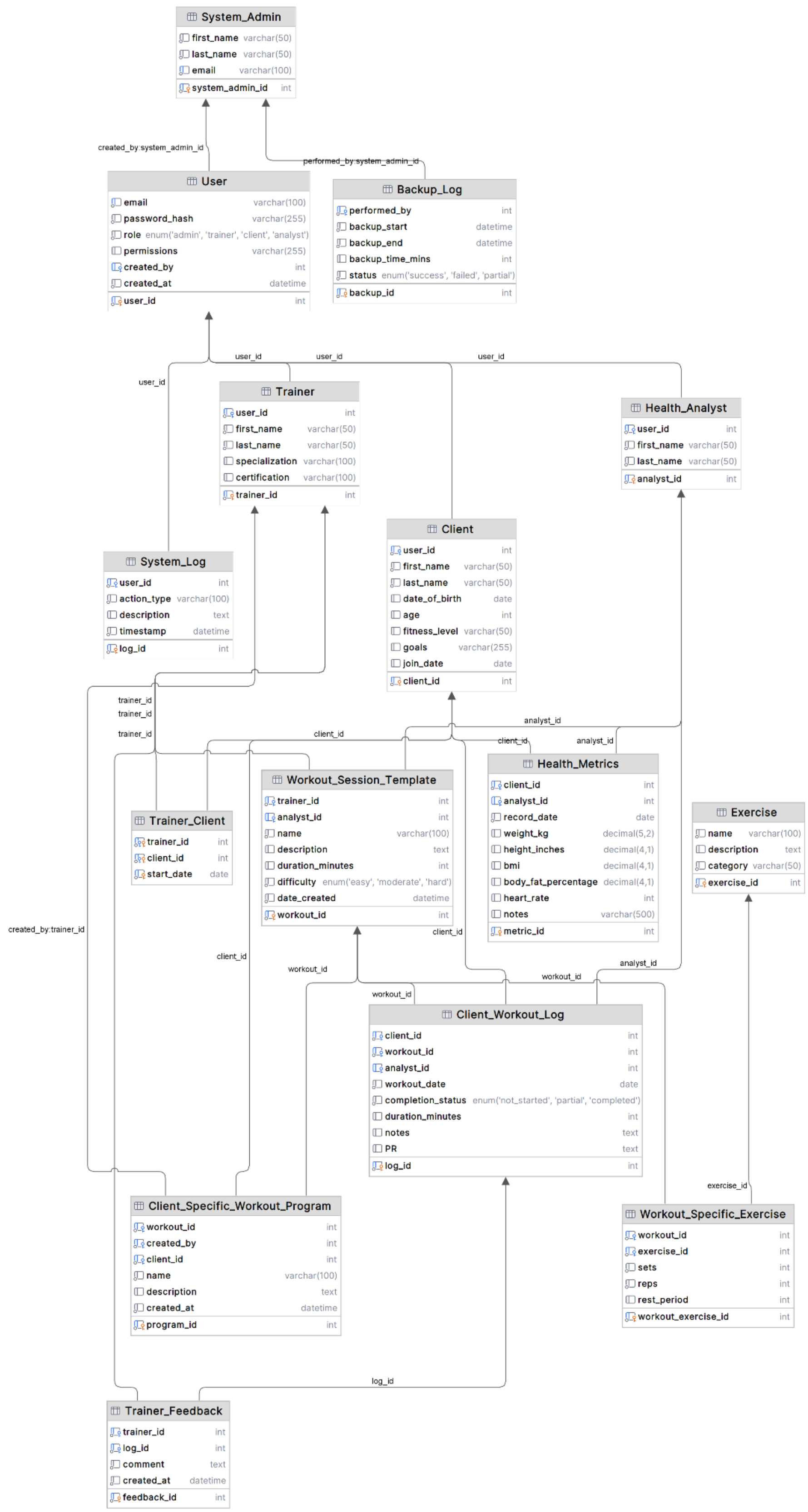
[Click Here to View Live on Lucid App]

Relational DB Diagram:



[Click Here to View Live on Lucid App]

Datagrip Database Diagram



Queries for User Stories

For each of the user stories, SQL SELECT, INSERT, UPDATE, or DELETE statements that would solve the data needs of the user story. Each Query labelled with its corresponding User Story Number (e.g., "2.4" for user story 4 of persona #2, or something similar).

1.1) As a system administrator, I want to add a user for a trainer profile that is verified with certifications so that new trainers can start using the platform immediately.

```
INSERT INTO User (email, password_hash, role, permissions)
VALUES (
   'marcus.rod@gmail.com',
   'marcdagoat25',
   'trainer',
   'create_exercise, edit_exercise, create_template, edit_template, create_program,
edit_program'
);

INSERT INTO Trainer (trainer_id, first_name, last_name, certification, specialization)
VALUES (LAST_INSERT_ID(), 'Marcus', 'Rodriguez', 'NASM-CPT', 'Strength Training');
```

1.2) As a system administrator, I want to add a user for a client profile so that new trainees can access their workouts right away.

```
INSERT INTO User (email, password_hash, role, permissions)
VALUES (
   'chestone@gmail.com',
   'chester2006',
   'client',
   'add_log, edit_log, view_exercises'
);

INSERT INTO Client (user_id, first_name, last_name, age, fitness_level)
VALUES (LAST_INSERT_ID(), 'Chester', 'Stone', 20, 'Intermediate');
```

1.3) As a system administrator, I want to remove outdated, or incorrect exercises so that the platform stays clean and organized.

```
DELETE FROM Exercise
WHERE exercise_id = 15;
```

1.4) As a system administrator, I want to manage all user accounts/profiles and permissions so that each role has the correct level of access to the platform's features.

```
UPDATE User
SET permissions = 'read_only'
```

```
WHERE user_id = (
   SELECT user_id FROM Client WHERE client_id = 8);
```

1.5) As a system administrator, I want to review flagged exercises or reported issues so that inappropriate or incorrect content is corrected or removed.

```
SELECT log_id, user_id, description, timestamp
FROM System_Log
WHERE action_type = 'FLAG_EXERCISE'
ORDER BY timestamp DESC;
```

1.6) As a system administrator, I want to monitor system logs and performance dashboards so that issues can be identified and resolved before they affect users.

```
SELECT *
FROM System_Log
ORDER BY timestamp DESC;
```

1.7) As a system administrator, I want the database to log when backups were last performed and notify me when a new one is due, so that data safety is maintained consistently.

```
-- Get most recent backup performed
SELECT *
FROM Backup_Log
WHERE status = 'SUCCESS'
ORDER BY backup_end DESC;

-- Log backup as performed
INSERT INTO Backup_Log (backup_start, status, performed_by)
VALUES (NOW(), 'IN_PROGRESS', 1);
UPDATE Backup_Log
SET backup_end = NOW(),
   status = 'SUCCESS'
WHERE backup_id = 'IN_PROGRESS';
```

2.1) As a personal trainer, I need to create workout programs with exercises, sets, reps, and rest periods that I can reuse and create effective programs for each client. (As seen on Wireframe1)

```
INSERT INTO Workout_Session_Template (trainer_id, name, description,
duration_minuted,difficulty)
VALUES (003, 'Beginner Strength Building', 'Intro program to develop proper form,
build core stability, and build foundational strength',45, 'easy');

INSERT INTO Workout_Specific_Exercise (workout_exercise_id, workout_id, exercise_id,
sets, reps, rest_period)
VALUES
(301, 23, 12, 3, 10, 60),
(302, 23, 8, 4, 12, 90),
```

```
(303, 23, 3, 3, 30, 30);
```

2.2) As a personal trainer, I need to organize and assign specific workout programs for clients so that they know exactly what to do each week. (As seen on Wireframe2)

```
INSERT INTO Client_Specific_Workout_Program (workout_id, created_by, client_id, name, description)
VALUES (12, 'Trainer' , 3, 'Sams_Program','Upper Body Week 1');
```

2.3) As a personal trainer, I need to view which clients have completed their assigned workouts so that I can identify who needs follow-up or motivation. (As seen on Wireframe1)

```
SELECT *,
FROM Client_Workout_Log,
WHERE completion_status = 'completed',
ORDER BY workout_date DESC ;
```

2.4) As a personal trainer, I need to see client progress metrics like consistency rate or personal records so that I can adjust their programs appropriately in the case it's too difficult or they need more challenges. (As seen on Wireframe1)

```
SELECT client_id,
COUNT(*) AS total_workouts,
SUM(completion_status = 'completed') AS completed,
ROUND(SUM(completion_status='completed') / COUNT(*), 2) AS completion_rate
FROM Client_Workout_Log
WHERE client_id = 25
```

2.5) As a personal trainer, I need to review clients' updated personal records so that I can adjust their future workout programs appropriately.   (As seen on wireframe2)

```
SELECT c.first_name, c.last_name, log.PR, log.completion_status, log.workout_date
FROM Client_Workout_Log log
JOIN Client c ON c.client_id = log.client_id
WHERE c.client_id = 25;
ORDER BY log.workout_date DESC;
```

2.6) As a personal trainer, I need to be able to leave notes/feedback on client workout logs so that I can provide encouragement or form corrections or adjust for any injuries, changes in health, or changing goals. (As seen on Wireframe2)

```
INSERT INTO Trainer_Feedback (log_id, trainer_id, comment)
VALUES (1, 1, 'great work!')
```

3.1) As a health analyst, I need the workout duration of all members as an average based on their completion. (As seen on Wireframe 5)

```
SELECT
  c.client_id,
  YEAR(cwl.date) AS year,
  WEEK(cwl.date, 1) AS week,
  AVG(cwl.duration_minutes) AS avg_duration
FROM Client c JOIN Client_Workout_Log cwl ON c.client_id = cwl.client_id
WHERE cwl.completion_status = 'completed'
GROUP BY c.client_id, YEAR(cwl.date), WEEK(cwl.workout_date, 1)
ORDER BY c.client_id, year, week;
```

3.2) As a health analyst, I need member background information including demographics, fitness levels, and goals. (As seen on Wireframe 6)

```
SELECT
 c.client_id,
 c.first_name,
 c.last_name,
 c.date_of_birth,
 c.Goals,
 c.fitness_level,
 c.Age,
 c.join_date
FROM Client c;
```

3.3) As a health analyst, I need each client's most recent health metric record to track their latest status. (As seen on Wireframe 6)

```
SELECT
  hm.client_id,
  hm.record_date,
  hm.weight_kg,
  hm.body_fat_percentage,
  hm.heart_rate
FROM Health_Metrics hm
WHERE hm.record_date = (
  SELECT MAX(record_date)
  FROM Health_Metrics
  WHERE client_id = hm.client_id
);
```

3.4) As a health analyst, I need analytics on health progression or decline based on weight and body fat percentage. (As seen on Wireframe 5)

```
SELECT
 client_id,
 MONTH(hm.record_date) AS month,
 AVG(weight_kg),
 AVG(body_fat_percentage)
FROM Health_Metrics hm
GROUP BY client_id, MONTH(hm.record_date)
```

```
ORDER BY client_id, month;
```

3.5) As a health analyst, I need workout completion rates related to each program in order to track succession analysis. (As seen on Wireframe 6)

```
SELECT
  cswp.program_id,
  cswp.name AS program_name,
  COUNT(cwl.log_id) AS total_workouts,
  SUM(cwl.completion_status = 'completed') AS completed_workouts,
  ROUND(
      SUM(cwl.completion_status = 'completed') / COUNT(cwl.log_id),
      3
  ) AS completion_rate
FROM Client_Specific_Workout_Program cswp JOIN Client_Workout_Log cwl
  ON cswp.workout_id = cwl.workout_id
  AND cswp.client_id = cwl.client_id
GROUP BY cswp.program_id, cswp.name
ORDER BY completion_rate DESC;
```

3.6) As a health analyst, I need to find which workout templates are most frequently used by the clients. (As seen on Wireframe 5)

```
SELECT
  cwl.workout_id,
  wst.name,
  COUNT(*) AS used
FROM Client_Workout_Log cwl JOIN Workout_Session_Template wst
  ON cwl.workout_id = wst.workout_id
GROUP BY cwl.workout_id, wst.name
ORDER BY used DESC;
```

4.1) As a client, I want to log my completed workouts with duration and notes so that I can track what I accomplished each session. (Wireframe 8)

```
INSERT INTO Client_Workout_Log (client_id,
                                workout_id,
                                date,
                                completion_status,
                                duration_minutes,
                                notes)
VALUES (1,
      5,
      '2025-11-23',
      'completed',
      62,
      'Felt strong today. Benched heavier than last week. Right shoulder a bit
tight.');
```

4.2) As a client, I want to view my most recently completed workouts so that I can see which sessions I've done and stay motivated. (Wireframe 7)

```
SELECT cwl.date AS workout_date,
     wst.name AS workout_name,
     cwl.completion_status,
     cwl.duration_minutes
FROM Client_Workout_Log cwl
       JOIN Workout_Session_Template wst ON cwl.workout_id = wst.workout_id
WHERE cwl.client_id = 1
AND cwl.completion_status = 'completed'
ORDER BY cwl.date DESC
LIMIT 10;
```

4.3) As a client, I want to view my coach-assigned weekly program with all exercise details so that I know exactly what to do each training session (Wireframe 8)

```
SELECT cswp.name AS program_name,
     e.name    AS exercise_name,
     wse.sets,
     wse.reps,
     wse.rest_period
FROM Client_Specific_Workout_Program cswp
       JOIN Workout_Session_Template wst ON cswp.workout_id = wst.workout_id
       JOIN Workout_Specific_Exercise wse ON wst.workout_id = wse.workout_id
       JOIN Exercise e ON wse.exercise_id = e.exercise_id
WHERE cswp.client_id = 1
ORDER BY wse.workout_exercise_id;
```

4.4) As a client, I want to see how many workouts I've completed this month compared to last month so that I can track if my consistency is improving. (Wireframe 7)

```
SELECT MONTH(date) AS month,
     COUNT(*)    AS workouts_completed
FROM Client_Workout_Log
WHERE client_id = 1
AND date >= '2025-10-01'
AND completion_status = 'completed'
GROUP BY MONTH(date)
ORDER BY month DESC;
```

4.5) As a client, I want to see how many personal records I've achieved this month so that I can track my progress and celebrate my improvements.

```
SELECT COUNT(*) AS total_prs_this_month
FROM Client_Workout_Log
WHERE client_id = 1
```

```sql
AND PR IS NOT NULL
AND PR != ''  # counts all PR rows with any text and values
AND workout_date >= '2025-10-23';
```

4.6) As a client, I want to delete incomplete workout logs so that my workout history only shows actual training sessions I completed.

```sql
DELETE
FROM Client_Workout_Log
WHERE client_id = 1
AND completion_status = 'not_started';
```

SQL Code:

SQL DDL (and any other related SQL) below.

```
DROP DATABASE IF EXISTS fitflow;
CREATE DATABASE IF NOT EXISTS fitflow;
USE fitflow;

DROP TABLE IF EXISTS System_Admin;
CREATE TABLE System_Admin
(
  system_admin_id INT PRIMARY KEY,
  first_name VARCHAR(50) NOT NULL,
  last_name  VARCHAR(50) NOT NULL,
  email VARCHAR(100) NOT NULL
);

DROP TABLE IF EXISTS User;
CREATE TABLE User(
  user_id       INT PRIMARY KEY AUTO_INCREMENT,
  email         VARCHAR(100) UNIQUE NOT NULL,
  password_hash VARCHAR(255)        NOT NULL,
  role          ENUM ('admin','trainer','client','analyst') NOT NULL,
  permissions   VARCHAR(255),
  created_by    INT NULL,
  created_at    DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT fk_user_created_by
      FOREIGN KEY (created_by)
      REFERENCES System_Admin(system_admin_id)
      ON DELETE SET NULL
      ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Health_Analyst;
CREATE TABLE Health_Analyst(
  analyst_id INT PRIMARY KEY AUTO_INCREMENT,
  user_id    INT NOT NULL,
  first_name VARCHAR(50) NOT NULL,
  last_name  VARCHAR(50) NOT NULL,
  CONSTRAINT fk_analyst_user
     FOREIGN KEY (user_id) REFERENCES User(user_id)
     ON DELETE CASCADE ON UPDATE CASCADE,

  UNIQUE (user_id)
);

DROP TABLE IF EXISTS Trainer;
CREATE TABLE Trainer(
  trainer_id     INT PRIMARY KEY AUTO_INCREMENT,
  user_id        INT NOT NULL,
  first_name     VARCHAR(50) NOT NULL,
  last_name      VARCHAR(50) NOT NULL,
  specialization VARCHAR(100),
```

```sql
  certification VARCHAR(100),

  CONSTRAINT fk_trainer_user
     FOREIGN KEY (user_id) REFERENCES User(user_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  UNIQUE (user_id)
);

DROP TABLE IF EXISTS Client;
CREATE TABLE Client(
  client_id     INT PRIMARY KEY AUTO_INCREMENT,
  user_id       INT NOT NULL,
  first_name    VARCHAR(50) NOT NULL,
  last_name     VARCHAR(50) NOT NULL,
  date_of_birth DATE,
  age           INT, -- derived attribute from date_of_birth
  fitness_level VARCHAR(50),
  goals         VARCHAR(255),
  join_date     DATE,
  CONSTRAINT fk_client_user
     FOREIGN KEY (user_id) REFERENCES User(user_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  UNIQUE (user_id)
);

DROP TABLE IF EXISTS System_Log;
CREATE TABLE System_Log(
  log_id      INT PRIMARY KEY AUTO_INCREMENT,
  user_id     INT NOT NULL,
  action_type VARCHAR(100) NOT NULL,
  description TEXT,
  timestamp   DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT fk_systemlog_user
     FOREIGN KEY (user_id) REFERENCES User(user_id)
     ON DELETE CASCADE ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Backup_Log;
CREATE TABLE Backup_Log(
  backup_id        INT PRIMARY KEY AUTO_INCREMENT,
  performed_by     INT NOT NULL,    -- system_admin_id
  backup_start     DATETIME NOT NULL,
  backup_end       DATETIME NOT NULL,
  backup_time_mins INT,
  status           ENUM('success','failed','partial') NOT NULL,
  CONSTRAINT fk_backuplog_admin
     FOREIGN KEY (performed_by) REFERENCES System_Admin(system_admin_id)
     ON DELETE CASCADE ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Health_Metrics;
CREATE TABLE Health_Metrics(
  metric_id           INT PRIMARY KEY AUTO_INCREMENT,
```

```
  client_id           INT NOT NULL,
  analyst_id          INT,
  record_date         DATE NOT NULL,
  weight_kg           DECIMAL(5,2),
  height_inches       DECIMAL(4,1),
  bmi                 DECIMAL(4,1),
  body_fat_percentage DECIMAL(4,1),
  heart_rate          INT,
  notes               VARCHAR(500),
  CONSTRAINT fk_metrics_client
     FOREIGN KEY (client_id) REFERENCES Client(client_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_metrics_analyst
     FOREIGN KEY (analyst_id) REFERENCES Health_Analyst(analyst_id)
     ON DELETE SET NULL ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Trainer_Client;
CREATE TABLE Trainer_Client(
  trainer_id INT NOT NULL,
  client_id  INT NOT NULL,
  start_date DATE NOT NULL,
  PRIMARY KEY (trainer_id, client_id, start_date),
  CONSTRAINT fk_tc_trainer
     FOREIGN KEY (trainer_id) REFERENCES Trainer(trainer_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_tc_client
     FOREIGN KEY (client_id) REFERENCES Client(client_id)
     ON DELETE CASCADE ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Exercise;
CREATE TABLE Exercise(
  exercise_id INT PRIMARY KEY AUTO_INCREMENT,
  name        VARCHAR(100) NOT NULL,
  description TEXT,
  category    VARCHAR(50) NOT NULL
);

DROP TABLE IF EXISTS Workout_Session_Template;
CREATE TABLE Workout_Session_Template(
  workout_id        INT PRIMARY KEY AUTO_INCREMENT,
  trainer_id        INT NOT NULL,
  analyst_id        INT,
  name              VARCHAR(100) NOT NULL,
  description       TEXT,
  duration_minutes  INT,
  difficulty        ENUM('easy','moderate','hard') NOT NULL,
  date_created      DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT fk_wst_trainer
     FOREIGN KEY (trainer_id) REFERENCES Trainer(trainer_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_wst_analyst
     FOREIGN KEY (analyst_id) REFERENCES Health_Analyst(analyst_id)
```

```
    ON DELETE SET NULL ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Workout_Specific_Exercise;
CREATE TABLE Workout_Specific_Exercise(
  workout_exercise_id INT PRIMARY KEY AUTO_INCREMENT,
  workout_id          INT NOT NULL,
  exercise_id         INT NOT NULL,
  sets                INT NOT NULL,
  reps                INT NOT NULL,
  rest_period         INT,
  CONSTRAINT fk_wse_workout
    FOREIGN KEY (workout_id) REFERENCES Workout_Session_Template(workout_id)
    ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_wse_exercise
    FOREIGN KEY (exercise_id) REFERENCES Exercise(exercise_id)
    ON DELETE RESTRICT ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Client_Specific_Workout_Program;
CREATE TABLE Client_Specific_Workout_Program(
  program_id  INT PRIMARY KEY AUTO_INCREMENT,
  workout_id  INT NOT NULL,
  created_by  INT NOT NULL,   -- trainer_id
  client_id   INT NOT NULL,
  name        VARCHAR(100) NOT NULL,
  description TEXT,
  created_at  DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT fk_cswp_workout
    FOREIGN KEY (workout_id) REFERENCES Workout_Session_Template(workout_id)
    ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_cswp_trainer
    FOREIGN KEY (created_by) REFERENCES Trainer(trainer_id)
    ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_cswp_client
    FOREIGN KEY (client_id) REFERENCES Client(client_id)
    ON DELETE CASCADE ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Client_Workout_Log;
CREATE TABLE Client_Workout_Log (
  log_id            INT PRIMARY KEY AUTO_INCREMENT,
  client_id         INT NOT NULL,
  workout_id        INT NOT NULL,  -- from Workout_Session_Template
  analyst_id        INT,
  workout_date      DATE NOT NULL,
  completion_status ENUM('not_started','partial','completed') NOT NULL DEFAULT
'not_started',
  duration_minutes  INT,
  notes             TEXT,
  PR                TEXT,     -- multivalued attribute
  CONSTRAINT fk_cwl_client
    FOREIGN KEY (client_id) REFERENCES Client(client_id)
    ON DELETE CASCADE ON UPDATE CASCADE,
```

```sql
  CONSTRAINT fk_cwl_workout
     FOREIGN KEY (workout_id) REFERENCES Workout_Session_Template(workout_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_cwl_analyst
     FOREIGN KEY (analyst_id) REFERENCES Health_Analyst(analyst_id)
     ON DELETE SET NULL ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Trainer_Feedback;
CREATE TABLE Trainer_Feedback(
  feedback_id INT PRIMARY KEY AUTO_INCREMENT,
  trainer_id  INT NOT NULL,
  log_id      INT NOT NULL,
  comment     TEXT NOT NULL,
  created_at  DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT fk_tf_trainer
     FOREIGN KEY (trainer_id) REFERENCES Trainer(trainer_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_tf_log
     FOREIGN KEY (log_id) REFERENCES Client_Workout_Log(log_id)
     ON DELETE CASCADE ON UPDATE CASCADE
);

-- INSERT STATEMENTS FOR SAMPLE DATA

-- SYSTEM ADMIN
INSERT INTO System_Admin (system_admin_id, first_name, last_name, email)
VALUES (001, 'Ava', 'Martinez', 'avamartinez@fitflow.com');

-- Note: All user_ids and role_ids are auto-incremented
-- TRAINER USER #1
INSERT INTO User (email, password_hash, role, permissions, created_by)
VALUES (
   'marcus.rod@fitflow.com',
   'marcdagoat2025',
   'trainer',
   'create_exercise, edit_exercise, create_template, edit_template, create_program,
edit_program',
   1
);

INSERT INTO Trainer (user_id, first_name, last_name, certification, specialization)
VALUES (LAST_INSERT_ID(), 'Marcus', 'Rodriguez', 'NASM-CPT', 'Strength Training');

-- TRAINER USER #2
INSERT INTO User (email, password_hash, role, permissions, created_by)
VALUES (
   'chad.zibz@gmail.com',
   'chaddddd387',
   'trainer',
   'create_exercise, edit_exercise, create_template, edit_template, create_program,
edit_program',
   1
);
```

```sql
INSERT INTO Trainer (user_id, first_name, last_name, certification, specialization)
VALUES (LAST_INSERT_ID(), 'Chad', 'Zibermann', 'ACE-CPT', 'Cardio & HIIT');

-- CLIENT USER #1
INSERT INTO User (email, password_hash, role, permissions, created_by)
VALUES (
   'Chester.stone@gmail.com',
   'gmfd25!',
   'client',
   'add_log, edit_log, view_exercises',
   1
);

INSERT INTO Client (user_id, first_name, last_name, date_of_birth, fitness_level, join_date,
goals)
VALUES (LAST_INSERT_ID(), 'Chester', 'Stone', '2001-11-02', 'Beginner', CURRENT_DATE,
'Build Muscle');

-- CLIENT USER #2
INSERT INTO User (email, password_hash, role, permissions, created_by)
VALUES (
   'CooperFlagg23@gmail.com',
   'bullCity',
   'client',
   'add_log, edit_log, view_exercises',
   1
);

INSERT INTO Client (user_id, first_name, last_name, date_of_birth, fitness_level, join_date,
goals)
VALUES (LAST_INSERT_ID(), 'Cooper', 'Flagg', '1998-01-15', 'Intermediate', CURRENT_DATE,
'General Fitness');

-- HEALTH ANALYST USER (user_id = 6)
INSERT INTO User (email, password_hash, role, permissions, created_by)
VALUES (
   'k.smith@fitflow.com',
   'pw6',
   'analyst',
   'analyze_logs, analyze_templates',
   1
);

INSERT INTO Health_Analyst (user_id, first_name, last_name)
VALUES (LAST_INSERT_ID(), 'Kennedy', 'Smith');

INSERT INTO Exercise (name, description, category)
VALUES ('Push-Up', 'Bodyweight chest and triceps exercise', 'Upper Body');

INSERT INTO Exercise (name, description, category)
VALUES ('Squat', 'Lower-body strength exercise', 'Legs');

INSERT INTO Exercise (name, description, category)
```

```sql
VALUES ('Plank', 'Core stabilization exercise', 'Core');

INSERT INTO Workout_Session_Template (trainer_id, analyst_id, name, description,
duration_minutes, difficulty)
VALUES (1, 1, 'Upper Body Blast', 'Push-oriented strength routine', 45, 'moderate');

INSERT INTO Workout_Session_Template (trainer_id, analyst_id, name, description,
duration_minutes, difficulty)
VALUES (2, 1, 'Leg Day Starter', 'Lower-body compound movements', 40, 'easy');

INSERT INTO Workout_Specific_Exercise (workout_id, exercise_id, sets, reps, rest_period)
VALUES (1, 1, 3, 12, 60);

INSERT INTO Workout_Specific_Exercise (workout_id, exercise_id, sets, reps, rest_period)
VALUES (2, 2, 4, 10, 90);

INSERT INTO Client_Specific_Workout_Program (workout_id, created_by, client_id, name,
description)
VALUES (1, 1, 1, 'Emily Strength Program', '4-week upper-body strength plan');

INSERT INTO Client_Specific_Workout_Program (workout_id, created_by, client_id, name,
description)
VALUES (2, 2, 2, 'Max Leg Program', 'Leg-focused starter program');

INSERT INTO Client_Workout_Log (client_id, workout_id, analyst_id, workout_date,
completion_status, duration_minutes, notes, PR)
VALUES (1, 1, 1, '2025-02-24', 'completed', 45, 'Felt great', 'Push-up:20');

INSERT INTO Client_Workout_Log (client_id, workout_id, analyst_id, workout_date,
completion_status, duration_minutes, notes, PR)
VALUES (2, 2, 1, '2025-02-24', 'partial', 30, 'Legs were tired', 'Squat:120lbs');

INSERT INTO Trainer_Feedback (trainer_id, log_id, comment)
VALUES (1, 1, 'Great job on form!');

INSERT INTO Trainer_Feedback (trainer_id, log_id, comment)
VALUES (2, 2, 'Increase warm-up next time.');

INSERT INTO System_Log (user_id, action_type, description)
VALUES (1, 'CREATE_USER', 'Admin created trainer profile for Marcus.');

INSERT INTO System_Log (user_id, action_type, description)
VALUES (2, 'CREATE_EXERCISE', 'Trainer Marcus added Squat exercise.');

INSERT INTO Backup_Log (performed_by, backup_start, backup_end, backup_time_mins, status)
VALUES (1, '2025-02-20 03:00:00', '2025-02-20 03:05:00', 5, 'success');

INSERT INTO Backup_Log (performed_by, backup_start, backup_end, backup_time_mins, status)
VALUES (1, '2025-02-22 03:00:00', '2025-02-22 03:08:00', 8, 'success');

SELECT *
FROM System_Admin;

SELECT *
```

```sql
FROM User;

SELECT *
FROM Health_Analyst;

SELECT *
FROM Trainer;

SELECT *
FROM Client;

SELECT *
FROM System_Log;

SELECT *
FROM Backup_Log;

SELECT *
FROM Health_Metrics;

SELECT *
FROM Trainer_Client;

SELECT *
FROM Exercise;

SELECT *
FROM Workout_Session_Template;

SELECT *
FROM Workout_Specific_Exercise;

SELECT *
FROM Client_Specific_Workout_Program;

SELECT *
FROM Client_Workout_Log;

SELECT *
FROM Trainer_Feedback;
```

REST API Matrix

- List all resources you've identified from the user stories.

Resource	GET	POST	PUT	DELETE
/exercise	Return full exercise library. **[Chester-3]**	n/a	n/a	n/a
/exercise/{ExerciseId}	n/a	n/a	Update a specific incorrect exercise's details. **[Ava-5]**	Remove an outdated/incorrect exercise. [**Ava-3]**
/workout_specific_exercise	Return all workout specific exercise **[Marcus-1]**	Create a new workout specific exercise **[Marcus-1]**	n/a	n/a
/workout_session_template	Return a list of all workout templates created by the trainer **[Marcus-2]** **[Chester-3]**	Create a new workout template **[Marcus-1]**	Update an existing workout template's info (name, difficulty, etc.) **[Marcus-1]**	Delete a workout template **[Marcus-1]**
/client_specific_workout_program	Get list of assigned programs **[Marcus-2]** Get the client's assigned workout program with template details **[Chester-3]**	Assign a workout template to client **[Marcus-2]**	Update assigned program **[Marcus-2]**	Remove an assigned program from a client **[Marcus-2]**
/client_workout_log	Return all logs where completion_status = TRUE (view who completed workouts) **[Marcus-3]** Return all workout logs for the authenticated client **[Chester-2]**	Create a new workout log entry with duration and notes **[Chester-1]**	Update a workout log's duration or notes **[Chester-1]**	Delete an entire workout session log **[Chester-6]**
/clients/{id}/progress	Return progress metrics for a client (completion rate, PR trends) **[Marcus-4]** **[Chester-5]**	n/a	n/a	n/a
/trainer_feedback	n/a	Create a feedback entry for a client's log **[Marcus-6]**	n/a	Delete feedback **[Marcus-6]**
/user	n/a	Create a new user account for profiles to be set up. Used for both trainers & clients. **[Ava-1], [Ava-2]**	n/a	n/a
/user/{id}	Return all details for a specific user. **[Ava-6]**	n/a	Update this specific user's permissions or status. **[Ava-4]**	n/a

/trainer	n/a	Create a Trainer profile after User creation. **[Ava-1]**	n/a	Delete trainer account. **[Ava-4]**
/client	n/a	Create a client profile after User creation. **[Ava-2]**	n/a	n/a
/system_logs	Return all system logs (admin monitoring). **[Ava-6]**	n/a	n/a	n/a
/system_logs{action_type}	Review system logs where the action_type is ‘flag_exercise’. **[Ava-5]**			
/backup_logs	Return last backup date + all backups. **[Ava-7]**	Insert new backup-log entry. **[Ava-7]**	n/a	n/a
/user/permissions	Return all roles’ permissions in the system. **[Ava-6]**	n/a	Update permissions for any user. **[Ava-4]**	n/a
/client/{id}	Returns all demographic info for a client **[Kennedy-2]**	n/a	Updates demographic info for a particular client **[Kennedy-2]**	n/a
/health_metrics	Returns all client health metrics **[Kennedy-3]**	Creates a new health metric profile for a client **[Kennedy-3]**	Updates a client’s health information **[Kennedy-3]**	n/a
/client_workout_log/avg_duration	Returns the average workout duration for all clients **[Kennedy-1]**	n/a	n/a	n/a
/client_specific_workout_program/{id}/completion_rate	Returns the average completion rate for a specific workout program **[Kennedy-5]**	n/a	n/a	n/a
/workout_session_template/used	Returns the amount of times each workout program has been used **[Kennedy-6]**	n/a	n/a	n/a
/health_metrics/{id}/month	Returns the health metrics from a range of months for a specific client **[Kennedy-4]**	n/a	n/a	n/a
/client_workout_log	Return all completed workout logs for client **[Chester-2]**	Create a new workout log entry with date, duration, completion_status, and notes **[Chester 1]**	Update a workout log's duration or notes **[Chester 1]**	Delete incomplete workout logs where completion_status = 'not_started' **[Chester-6]**
/client_workout_log/pr/monthly	Return count of personal records achieved this month **[Chester-5]**	n/a	n/a	n/a
/client_workout_log/completion_rate/monthly	Return workout completion count: current month vs previous month, where	n/a	n/a	n/a

	completion_status = 'completed' **[Chester-4]**			
/client_workout_log/{id}	Return details of a specific workout log including workout name **[Chester-2]**	n/a	n/a	n/a
/client_specific_workout_program/exercises	Return all exercises in the client's assigned program (program_name, exercise_name, sets, reps, rest_period) **[Chester-3]**		n/a	n/a

Finalized SQL Code:

```
DROP DATABASE IF EXISTS fitflow;
CREATE DATABASE IF NOT EXISTS fitflow;
USE fitflow;

DROP TABLE IF EXISTS System_Admin;
CREATE TABLE System_Admin
(
  system_admin_id INT PRIMARY KEY,
  first_name VARCHAR(50) NOT NULL,
  last_name  VARCHAR(50) NOT NULL,
  email VARCHAR(100) NOT NULL
);

DROP TABLE IF EXISTS User;
CREATE TABLE User(
  user_id       INT PRIMARY KEY AUTO_INCREMENT,
  email         VARCHAR(100) UNIQUE NOT NULL,
  password_hash VARCHAR(255)        NOT NULL,
  role          ENUM ('admin','trainer','client','analyst') NOT NULL,
  permissions   VARCHAR(255),
  created_by    INT NULL,
  created_at    DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT fk_user_created_by
      FOREIGN KEY (created_by)
      REFERENCES System_Admin(system_admin_id)
      ON DELETE SET NULL
      ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Health_Analyst;
CREATE TABLE Health_Analyst(
  analyst_id INT PRIMARY KEY AUTO_INCREMENT,
  user_id    INT NOT NULL,
  first_name VARCHAR(50) NOT NULL,
  last_name  VARCHAR(50) NOT NULL,
  CONSTRAINT fk_analyst_user
     FOREIGN KEY (user_id) REFERENCES User(user_id)
     ON DELETE CASCADE ON UPDATE CASCADE,

  UNIQUE (user_id)
);

DROP TABLE IF EXISTS Trainer;
CREATE TABLE Trainer(
  trainer_id    INT PRIMARY KEY AUTO_INCREMENT,
  user_id       INT NOT NULL,
  first_name    VARCHAR(50) NOT NULL,
  last_name     VARCHAR(50) NOT NULL,
  specialization VARCHAR(100),
  certification VARCHAR(100),
```

```sql
  CONSTRAINT fk_trainer_user
     FOREIGN KEY (user_id) REFERENCES User(user_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  UNIQUE (user_id)
);

DROP TABLE IF EXISTS Client;
CREATE TABLE Client(
  client_id     INT PRIMARY KEY AUTO_INCREMENT,
  user_id       INT NOT NULL,
  first_name    VARCHAR(50) NOT NULL,
  last_name     VARCHAR(50) NOT NULL,
  date_of_birth DATE,
  age           INT, -- derived attribute from date_of_birth
  fitness_level VARCHAR(50),
  goals         VARCHAR(255),
  join_date     DATE,
  CONSTRAINT fk_client_user
     FOREIGN KEY (user_id) REFERENCES User(user_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  UNIQUE (user_id)
);

DROP TABLE IF EXISTS System_Log;
CREATE TABLE System_Log(
  log_id      INT PRIMARY KEY AUTO_INCREMENT,
  user_id     INT NOT NULL,
  action_type VARCHAR(100) NOT NULL,
  description TEXT,
  timestamp   DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT fk_systemlog_user
     FOREIGN KEY (user_id) REFERENCES User(user_id)
     ON DELETE CASCADE ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Backup_Log;
CREATE TABLE Backup_Log(
  backup_id        INT PRIMARY KEY AUTO_INCREMENT,
  performed_by     INT NOT NULL,   -- system_admin_id
  backup_start     DATETIME NOT NULL,
  backup_end       DATETIME NOT NULL,
  backup_time_mins INT,
  status           ENUM('success','failed','partial') NOT NULL,
  CONSTRAINT fk_backuplog_admin
     FOREIGN KEY (performed_by) REFERENCES System_Admin(system_admin_id)
     ON DELETE CASCADE ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Health_Metrics;
CREATE TABLE Health_Metrics(
  metric_id           INT PRIMARY KEY AUTO_INCREMENT,
  client_id           INT NOT NULL,
  analyst_id          INT,
```

```
  record_date          DATE NOT NULL,
  weight_kg            DECIMAL(5,2),
  height_inches        DECIMAL(4,1),
  bmi                  DECIMAL(4,1),
  body_fat_percentage DECIMAL(4,1),
  heart_rate           INT,
  notes                VARCHAR(500),
  CONSTRAINT fk_metrics_client
     FOREIGN KEY (client_id) REFERENCES Client(client_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_metrics_analyst
     FOREIGN KEY (analyst_id) REFERENCES Health_Analyst(analyst_id)
     ON DELETE SET NULL ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Trainer_Client;
CREATE TABLE Trainer_Client(
  trainer_id INT NOT NULL,
  client_id  INT NOT NULL,
  start_date DATE NOT NULL,
  PRIMARY KEY (trainer_id, client_id, start_date),
  CONSTRAINT fk_tc_trainer
     FOREIGN KEY (trainer_id) REFERENCES Trainer(trainer_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_tc_client
     FOREIGN KEY (client_id) REFERENCES Client(client_id)
     ON DELETE CASCADE ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Exercise;
CREATE TABLE Exercise(
  exercise_id INT PRIMARY KEY AUTO_INCREMENT,
  name        VARCHAR(100) NOT NULL,
  description TEXT,
  category    VARCHAR(50) NOT NULL
);

DROP TABLE IF EXISTS Workout_Session_Template;
CREATE TABLE Workout_Session_Template(
  workout_id        INT PRIMARY KEY AUTO_INCREMENT,
  trainer_id        INT NOT NULL,
  analyst_id        INT,
  name              VARCHAR(100) NOT NULL,
  description       TEXT,
  duration_minutes  INT,
  difficulty        ENUM('easy','moderate','hard') NOT NULL,
  date_created      DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT fk_wst_trainer
     FOREIGN KEY (trainer_id) REFERENCES Trainer(trainer_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_wst_analyst
     FOREIGN KEY (analyst_id) REFERENCES Health_Analyst(analyst_id)
     ON DELETE SET NULL ON UPDATE CASCADE
);
```

```sql
DROP TABLE IF EXISTS Workout_Specific_Exercise;
CREATE TABLE Workout_Specific_Exercise(
  workout_exercise_id INT PRIMARY KEY AUTO_INCREMENT,
  workout_id          INT NOT NULL,
  exercise_id         INT NOT NULL,
  sets                INT NOT NULL,
  reps                INT NOT NULL,
  rest_period         INT,
  CONSTRAINT fk_wse_workout
     FOREIGN KEY (workout_id) REFERENCES Workout_Session_Template(workout_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_wse_exercise
     FOREIGN KEY (exercise_id) REFERENCES Exercise(exercise_id)
     ON DELETE RESTRICT ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Client_Specific_Workout_Program;
CREATE TABLE Client_Specific_Workout_Program(
  program_id  INT PRIMARY KEY AUTO_INCREMENT,
  workout_id  INT NOT NULL,
  created_by  INT NOT NULL,   -- trainer_id
  client_id   INT NOT NULL,
  name        VARCHAR(100) NOT NULL,
  description TEXT,
  created_at  DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT fk_cswp_workout
     FOREIGN KEY (workout_id) REFERENCES Workout_Session_Template(workout_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_cswp_trainer
     FOREIGN KEY (created_by) REFERENCES Trainer(trainer_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_cswp_client
     FOREIGN KEY (client_id) REFERENCES Client(client_id)
     ON DELETE CASCADE ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Client_Workout_Log;
CREATE TABLE Client_Workout_Log (
  log_id            INT PRIMARY KEY AUTO_INCREMENT,
  client_id         INT NOT NULL,
  workout_id        INT NOT NULL,   -- from Workout_Session_Template
  analyst_id        INT,
  workout_date      DATE NOT NULL,
  completion_status ENUM('not_started','partial','completed') NOT NULL DEFAULT
'not_started',
  duration_minutes  INT,
  notes             TEXT,
  PR                TEXT,     -- multivalued attribute
  CONSTRAINT fk_cwl_client
     FOREIGN KEY (client_id) REFERENCES Client(client_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_cwl_workout
     FOREIGN KEY (workout_id) REFERENCES Workout_Session_Template(workout_id)
```

```sql
     ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_cwl_analyst
     FOREIGN KEY (analyst_id) REFERENCES Health_Analyst(analyst_id)
     ON DELETE SET NULL ON UPDATE CASCADE
);

DROP TABLE IF EXISTS Trainer_Feedback;
CREATE TABLE Trainer_Feedback(
  feedback_id INT PRIMARY KEY AUTO_INCREMENT,
  trainer_id  INT NOT NULL,
  log_id      INT NOT NULL,
  comment     TEXT NOT NULL,
  created_at  DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT fk_tf_trainer
     FOREIGN KEY (trainer_id) REFERENCES Trainer(trainer_id)
     ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT fk_tf_log
     FOREIGN KEY (log_id) REFERENCES Client_Workout_Log(log_id)
     ON DELETE CASCADE ON UPDATE CASCADE
);

-- INSERT STATEMENTS FOR SAMPLE DATA

-- SYSTEM ADMIN
INSERT INTO System_Admin (system_admin_id, first_name, last_name, email)
VALUES (001, 'Ava', 'Martinez', 'avamartinez@fitflow.com');

-- Note: All user_ids and role_ids are auto-incremented
-- TRAINER USER #1
INSERT INTO User (email, password_hash, role, permissions, created_by)
VALUES (
   'marcus.rod@fitflow.com',
   'marcdagoat2025',
   'trainer',
   'create_exercise, edit_exercise, create_template, edit_template, create_program,
edit_program',
   1
);

INSERT INTO Trainer (user_id, first_name, last_name, certification, specialization)
VALUES (LAST_INSERT_ID(), 'Marcus', 'Rodriguez', 'NASM-CPT', 'Strength Training');

-- TRAINER USER #2
INSERT INTO User (email, password_hash, role, permissions, created_by)
VALUES (
   'chad.zibz@gmail.com',
   'chaddddd387',
   'trainer',
   'create_exercise, edit_exercise, create_template, edit_template, create_program,
edit_program',
   1
);

INSERT INTO Trainer (user_id, first_name, last_name, certification, specialization)
```

```sql
VALUES (LAST_INSERT_ID(), 'Chad', 'Zibermann', 'ACE-CPT', 'Cardio & HIIT');

-- CLIENT USER #1
INSERT INTO User (email, password_hash, role, permissions, created_by)
VALUES (
   'Chester.stone@gmail.com',
   'gmfd25!',
   'client',
   'add_log, edit_log, view_exercises',
   1
);

INSERT INTO Client (user_id, first_name, last_name, date_of_birth, fitness_level, join_date,
goals)
VALUES (LAST_INSERT_ID(), 'Chester', 'Stone', '2001-11-02', 'Beginner', CURRENT_DATE,
'Build Muscle');

-- CLIENT USER #2
INSERT INTO User (email, password_hash, role, permissions, created_by)
VALUES (
   'CooperFlagg23@gmail.com',
   'bullCity',
   'client',
   'add_log, edit_log, view_exercises',
   1
);

INSERT INTO Client (user_id, first_name, last_name, date_of_birth, fitness_level, join_date,
goals)
VALUES (LAST_INSERT_ID(), 'Cooper', 'Flagg', '1998-01-15', 'Intermediate', CURRENT_DATE,
'General Fitness');

-- HEALTH ANALYST USER (user_id = 6)
INSERT INTO User (email, password_hash, role, permissions, created_by)
VALUES (
   'k.smith@fitflow.com',
   'pw6',
   'analyst',
   'analyze_logs, analyze_templates',
   1
);

INSERT INTO Health_Analyst (user_id, first_name, last_name)
VALUES (LAST_INSERT_ID(), 'Kennedy', 'Smith');

INSERT INTO Exercise (name, description, category)
VALUES ('Push-Up', 'Bodyweight chest and triceps exercise', 'Upper Body');

INSERT INTO Exercise (name, description, category)
VALUES ('Squat', 'Lower-body strength exercise', 'Legs');

INSERT INTO Exercise (name, description, category)
VALUES ('Plank', 'Core stabilization exercise', 'Core');
```

```sql
INSERT INTO Workout_Session_Template (trainer_id, analyst_id, name, description,
duration_minutes, difficulty)
VALUES (1, 1, 'Upper Body Blast', 'Push-oriented strength routine', 45, 'moderate');

INSERT INTO Workout_Session_Template (trainer_id, analyst_id, name, description,
duration_minutes, difficulty)
VALUES (2, 1, 'Leg Day Starter', 'Lower-body compound movements', 40, 'easy');

INSERT INTO Workout_Specific_Exercise (workout_id, exercise_id, sets, reps, rest_period)
VALUES (1, 1, 3, 12, 60);

INSERT INTO Workout_Specific_Exercise (workout_id, exercise_id, sets, reps, rest_period)
VALUES (2, 2, 4, 10, 90);

INSERT INTO Client_Specific_Workout_Program (workout_id, created_by, client_id, name,
description)
VALUES (1, 1, 1, 'Emily Strength Program', '4-week upper-body strength plan');

INSERT INTO Client_Specific_Workout_Program (workout_id, created_by, client_id, name,
description)
VALUES (2, 2, 2, 'Max Leg Program', 'Leg-focused starter program');

INSERT INTO Client_Workout_Log (client_id, workout_id, analyst_id, workout_date,
completion_status, duration_minutes, notes, PR)
VALUES (1, 1, 1, '2025-02-24', 'completed', 45, 'Felt great', 'Push-up:20');

INSERT INTO Client_Workout_Log (client_id, workout_id, analyst_id, workout_date,
completion_status, duration_minutes, notes, PR)
VALUES (2, 2, 1, '2025-02-24', 'partial', 30, 'Legs were tired', 'Squat:120lbs');

INSERT INTO Trainer_Feedback (trainer_id, log_id, comment)
VALUES (1, 1, 'Great job on form!');

INSERT INTO Trainer_Feedback (trainer_id, log_id, comment)
VALUES (2, 2, 'Increase warm-up next time.');

INSERT INTO System_Log (user_id, action_type, description)
VALUES (1, 'CREATE_USER', 'Admin created trainer profile for Marcus.');

INSERT INTO System_Log (user_id, action_type, description)
VALUES (2, 'CREATE_EXERCISE', 'Trainer Marcus added Squat exercise.');

INSERT INTO Backup_Log (performed_by, backup_start, backup_end, backup_time_mins, status)
VALUES (1, '2025-02-20 03:00:00', '2025-02-20 03:05:00', 5, 'success');

INSERT INTO Backup_Log (performed_by, backup_start, backup_end, backup_time_mins, status)
VALUES (1, '2025-02-22 03:00:00', '2025-02-22 03:08:00', 8, 'success');

SELECT *
FROM System_Admin;

SELECT *
FROM User;
```

```sql
SELECT *
FROM Health_Analyst;

SELECT *
FROM Trainer;

SELECT *
FROM Client;

SELECT *
FROM System_Log;

SELECT *
FROM Backup_Log;

SELECT *
FROM Health_Metrics;

SELECT *
FROM Trainer_Client;

SELECT *
FROM Exercise;

SELECT *
FROM Workout_Session_Template;

SELECT *
FROM Workout_Specific_Exercise;

SELECT *
FROM Client_Specific_Workout_Program;

SELECT *
FROM Client_Workout_Log;

SELECT *
FROM Trainer_Feedback;
```